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# Detecting AI Training in Datacentre Power

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## Abstract

Compute-governance regimes that restrict large training runs presuppose that a regulator can tell when training happens on monitored hardware, and a power meter outside the operator’s control is the one passive channel available against an operator who does not cooperate. We ask what such a meter can certify, and answer with a ladder of four claims, each with a stated information cost and a hard ceiling. Evidence comes from a literature-parameterized scenario model of datacentre AI workloads and an explicit off-chip observation model, against both a benign operator and a concealing one who knows the detector. The iteration cadence of synchronous training is recoverable by a tracked cyclostationary detector, but the complete pipeline admits no analytic calibration, so we report it as an effective score with per-trace surrogate calibration rather than a constant false-alarm-rate test. Our central result is the de-periodicisation frontier, which prices hiding against measured throughput anchors and is strongly asymmetric: varying real work per iteration hides from every fixed-frequency test at essentially zero cost, while the tracking class holds until that variation is extreme, where it still detects about half of runs. That extreme level is also free in throughput, so its price, if any, lies in learning efficiency, which we do not measure. Two limits are firmer than the detectors. The observation channel is itself a condition on every claim — a one-second-integrating, one-hertz meter defeats all detectors on an honest workload — and the meter certifies a physical schedule, not training semantics: five non-training loads, including one computing gradients and discarding them, are classified as training by every rule we test. Escaping that ceiling means leaving the passive channel for an active challenge protocol, which

we specify but do not run.

## 1. Introduction

AI governance agreements will require regulating training runs (Shavit, 2023; Sastry et al., 2024; Baker et al., 2025; Heim & Koessler, 2024). This will require an external verifier to identify when training is occurring on a monitored cluster, against an operator who is trying to hide their training. There are multiple channels that could provide the verifier with diagnostic information to identify hidden training. These include on-chip mechanisms that log activities for later analysis, network taps that monitor data exchanged between chips, or human-centred approaches such as whistleblower programs or interviews of operator personnel. However each of these channels requires trusting either hardware that is under the control of the operator, or the operator personnel themselves. Conversely, off-chip sensors that take analog, physical measurements of the cluster (e.g. power or electromagnetic emissions) do not require trusting the operator. Such sensors could be retrofitted to existing datacentres (e.g. clamping a power meter on an input cable) and secured against tampering by the operator.

AI training has already been shown to leave a signature in the power drawn by the hardware that can serve as a diagnostic to identify and separate training from inference. A 10 MHz current probe clamped around the GPU’s PCIe power supply cables can separate training from inference and from non-AI work at 97% accuracy (Gargiulo & Kulp, 2026), a result robust against many but not all adversarial strategies. On-chip 1 Hz telemetry counters that measure power (among other telemetry: utilisation, memory, etc.) can classify training at 95% accuracy, again a result robust against many but not all adversarial strategies (Rahman & Tajdari, 2026). In practice, any realistic AI governance agreements that use off-chip sensors will rarely have such fine-grained access to the GPU hardware. Power meters that can be retrofitted to existing facilities and secured against tampering are likely to sit at the level of the rack or facility input, with sampling frequencies  $\sim 10$  Hz (i.e. three to four orders of magnitude slower than the board probe) and behind a power-delivery path that low-passes and dilutes the signal. At this level training can be identified and it is large enough that datacentre and grid operators treat it as an engineering

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control problem to be dealt with (Patel et al., 2023; Ko & Zhu, 2025).

In this paper we develop a new method for detecting training signatures given noisy facility level power measurements, against an adversarial operator who is trying to hide their training.

We do:

- \* specify an off-chip observation model and a literature-parameterised generator of datacentre training and inference power,
- \* construct a detector for the cadence that tracks the wandering line, borrowing the technique from signal analysis and GW astrophysics
- \* We construct the Pareto frontier and show the trade-off between how much of the cadence a concealing operator removes and what that removal costs in measured throughput (Section 6).
- \* We derive a minimum meter specification, the sampling rate, integration time and in-band transfer i.e. the specs needed for a meter to work
- \* We restrict attention to synchronous, iteration-structured training observed through a time-resolved trace; the integrated-energy channel, asynchronous schedules and the calibration of the generator to distributed hardware are outside scope, and Sections 7 and 7.2 say what that costs.

Unlike a covert transmitter, a training operator cannot freely reshape its signal: the communication phase has a fixed physical duration, and every distortion of the cadence costs throughput. It is this constraint that makes a cost-of-hiding bound possible.

This paper is organised as follows. Section 2 gives the model of datacentre training and inference power that generates every trace below, and Section 3 fixes what the verifier sees through an off-chip meter and what a concealing operator may do to hide from it. Section 4 develops the detector, a Viterbi line tracker; Section 5 reports its power against the inference null and the minimum meter it needs; and Section 6 prices what it costs a concealing operator to defeat it. Section 7 states what a detection does and does not certify, holds the model against measured single-GPU hardware, and reads the results back as a ladder of claims a regulator can buy.

## 2. A model of datacentre training power

In this section we build a model of the power draw from a data centre running AI workloads. This model will serve as a synthetic data generator to test our methods, in lieu of having access to real empirical data. We adopt the stochastic model

of AI training power from Ko & Zhu (2025), developed for the analysis of grid stability, modified to add an Ornstein-Uhlenbeck drift on the training frequency. The reader is referred to Ko & Zhu (2025) for the full training model and its context.

This section is organised as follows. Section 2.1 describes the power draw from a training workload, following Ko & Zhu (2025), and adds our drift of its cadence. Section 2.2 describes the power draw from a fine-tuning workload. Section 2.3 describes the power draw from an inference workload. Section 2.4 combines these workloads into the load of a whole datacentre: one dominant workload, either training or inference, over a background of smaller trainings and fine-tunings. Table 1 collects every parameter value of the model.

### 2.1. Training

Multi GPU AI training of the kind that typifies frontier models is composed of a two-phase iterative cycle. First there is a power intensive, high compute “up” phase where the forward/backward pass happens. Second there is a low power, low compute, communication “down” phase where the gradient all reduce happens.

Iteration  $i$  has period

$$T_i = \frac{1}{f_0 (1 + \xi_i)}, \quad \xi_i \sim \mathcal{N}(0, \sigma_\xi^2), \quad (1)$$

where  $f_0$  is the baseline fluctuation frequency. For a given training run  $f_0$  is fixed. Between training runs  $f_0$  can vary depending on model size, batch size, parallelism layout, interconnect and hardware (Patel et al., 2024a; Jain et al., 2024). To describe the inter-workload variation, we take  $f_0 \sim \mathcal{U}(f_0^{\text{lo}}, f_0^{\text{hi}})$ . The period splits into an up-phase of duration  $\rho_i T_i$  and a down-phase of duration  $(1 - \rho_i) T_i$ , with the phase ratio  $\rho_i \sim \mathcal{U}(\rho^{\text{lo}}, \rho^{\text{hi}})$  redrawn each iteration. Writing  $t$  for time within iteration  $i$ ,  $P(t)$  for the power the run draws above the facility’s static overhead (added in Section 3.1), and  $P_{\text{peak}}$  for its peak,

$$\frac{P(t)}{P_{\text{peak}}} = \begin{cases} 1 + \Delta_i^{\text{up}} + \nu^{\text{up}}(t), & t < \rho_i T_i, \\ 1 - \Delta_i^{\text{dn}} + \nu^{\text{dn}}(t), & t \geq \rho_i T_i, \end{cases} \quad (2)$$

where the per-phase level offsets are  $\Delta_i^{\text{up}} \sim \mathcal{N}(0, \sigma_\Delta^2)$  and  $\Delta_i^{\text{dn}} \sim \mathcal{N}(\mu_\Delta, \sigma_\Delta^2)$ , and  $\nu^{\text{up}}, \nu^{\text{dn}}$  are zero-mean intra-phase Gaussian fluctuations with standard deviations  $\kappa_\nu s_i^{\text{up}}$  and  $\kappa_\nu s_i^{\text{dn}}$ . The  $s_i^{\text{up}}$  and  $s_i^{\text{dn}}$  are redrawn each phase, uniformly from the ranges  $[s^{\text{up,lo}}, s^{\text{up,hi}}]$  and  $[s^{\text{dn,lo}}, s^{\text{dn,hi}}]$ , and  $\kappa_\nu$  is a fixed scale factor. Table 1 lists the values of every parameter in this section.

Equations (1) and (2) are the exact training workload model from Ko & Zhu (2025). We make one modification, adding a slow drift to the cadence  $f_0$  itself. The jitter  $\xi_i$  does not

produce drift. Because it is redrawn independently each iteration, the cadence averaged over many iterations is still  $f_0$ . The timing errors accumulate as a random walk in phase, which broadens the spectral line into a Lorentzian of full width  $\approx 2\pi\sigma_\xi^2 f_0$  (to leading order in  $\sigma_\xi$ ) but leaves its centre at  $f_0$ . Real runs vary differently. Specifically, the iteration time changes in ways that persist. Every training step waits for the slowest GPU, so one slow component slows the whole run. In a production LLM cluster, these slowdowns last for the whole job rather than varying from step to step (Lin et al., 2025). Hardware can also keep running but degrade: CPU contention, overheating GPUs and congested network links cut throughput by tens of percent for periods of tens of seconds to hours (Wu et al., 2024). Over a day, temperature shifts throughput by a further 1–2% (Grattafiori et al., 2024).

We model this by letting the cadence of iteration  $i$  be  $f_0 g_i$ , where  $g_i$  is a discrete Ornstein–Uhlenbeck (first-order autoregressive) process that starts at  $g_0 = 1$  and reverts to 1:

$$g_i - 1 = (1 - \theta)(g_{i-1} - 1) + \epsilon_i, \quad \epsilon_i \sim \mathcal{N}(0, \sigma_g^2). \quad (3)$$

The period of Equation (1) becomes

$$T_i = \frac{1}{f_0 g_i (1 + \xi_i)}, \quad (4)$$

which reduces to Equation (1) when  $g_i \equiv 1$ . The stationary variance of Equation (3) is  $\sigma_g^2/(2\theta - \theta^2)$ , so we set

$$\sigma_g = \frac{\sigma_f}{f_0} \sqrt{2\theta - \theta^2}, \quad (5)$$

which makes the stationary standard deviation of the instantaneous cadence  $f_0 g_i$  equal to  $\sigma_f$ , in Hz. The autocorrelation of  $g_i$  at lag  $k$  is  $(1 - \theta)^k$ , so the cadence stays correlated over  $\sim 1/\theta$  iterations. Over that span it wanders by  $\sigma_f$ , which can exceed both the line width set by  $\xi_i$  and the frequency resolution of a trace. We clip  $g_i$  from below at  $g_{\min}$  so the period stays finite. We hold  $\theta$  fixed and treat  $\sigma_f$  as a free parameter, sweeping it in Section 5.

## 2.2. Fine-tuning

Fine-tuning workloads can be represented using the same structure as in Section 2.1, with different parameters (Table 1). We denote the fine-tuning frequency as  $f_1$ . Compared with training, it has a lower cadence band, a longer tail phase (larger  $\rho_i$ ), a deeper tail-to-idle swing (larger  $\mu_\Delta$ ), and slightly smaller level offsets and intra-phase noise. We apply the cadence drift of Equation (3) only to the dominant run; fine-tuning and background training runs keep a fixed cadence ( $\sigma_f = 0$ ).

## 2.3. Inference

Ko & Zhu (2025) do not present a model for power draw of inference workloads. We now construct one. We superpose three structures on a constant mean serving power load  $B$ . First, token generation (decode). Serving systems use continuous batching, which adds and removes requests from the batch at every step (Yu et al., 2022; Kwon et al., 2023). A per-step cadence exists, but the batch composition reshuffles every step. Unlike training, there is therefore no characteristic frequency: decode power smears across a broad band with no coherent frequency path. We model it as band-limited noise  $b(t)$  of unit RMS over  $[f_b^{\text{lo}}, f_b^{\text{hi}}]$ , scaled by an amplitude  $A$ . Second, expert-load imbalance. Under sparse mixture-of-experts routing, tokens are not spread evenly across the experts (Shazeer et al., 2017), so the work per step varies with the mix of requests. We model this as a slow envelope on the decode power: a unit-variance Ornstein–Uhlenbeck process  $u(t)$  with correlation time  $\tau_u$ , entering at depth  $m$ . Amplitude modulation spreads the decode energy into sidebands, which makes the null harder, not easier. Third, prompt processing (prefill). Each request begins with a prompt phase that is brief, compute-bound, and draws more power than token generation (Patel et al., 2024a;b). We model these phases as box pulses  $s(\cdot)$  of duration  $d_p$  and height  $a$ , arriving at times  $\tau_k$  from a Poisson process of rate  $\lambda_p$ . These memoryless arrivals carry no line and simply raise the broadband in-band level. Together,

$$\frac{P^{\text{inf}}(t)}{P_{\text{peak}}} = B + A [1 + m u(t)] b(t) + \sum_k a s(t - \tau_k). \quad (6)$$

The forms of these structures follow how LLM serving systems schedule work; their amplitudes are our choice (Table 1). We set them so that inference is a loud confound rather than a quiet one. As Figure 1 shows, it delivers more in-band power than training but an order of magnitude less spectral contrast.

## 2.4. Aggregate

A datacentre rarely runs one job at a time. Following Ko & Zhu (2025) we write the load behind a single meter as the aggregate of one dominant training run over a background of smaller trainings and fine-tunings,

$$P_{\text{AI}}(t) = P^{\text{main}}(t) + \sum_{j=1}^{N_{\text{tr}}} P_j^{\text{tr}}(t) + \sum_{k=1}^{N_{\text{ft}}} P_k^{\text{ft}}(t). \quad (7)$$

where  $P^{\text{main}}$  is the power of the dominant training run,  $P_j^{\text{tr}}$  that of the  $j$ -th of  $N_{\text{tr}}$  smaller training runs, and  $P_k^{\text{ft}}$  that of the  $k$ -th of  $N_{\text{ft}}$  fine-tuning runs. Each term is an independent draw of Equation (2), with its own cadence, phase ratios, and level offsets.  $P^{\text{main}}$  and the  $P_j^{\text{tr}}$  use the training parameters and the  $P_k^{\text{ft}}$  use the fine-tuning parameters; only  $P^{\text{main}}$

carries the cadence drift of Equation (3). Peak power is split between the dominant run, the small trainings, and the fine-tunings in the ratio  $w_{\text{main}} : w_{\text{tr}} : w_{\text{ft}}$ , and each background share is divided equally among its members. Each small training draws its cadence from the same band as the dominant run, so the background carries weak lines of its own inside the search band.

We define the null aggregate as the same datacentre with the dominant training run replaced by inference (Section 2.3). The background is unchanged,

$$P_{\text{AI}}^{\text{null}}(t) = P^{\text{inf}}(t) + \sum_{j=1}^{N_{\text{tr}}} P_j^{\text{tr}}(t) + \sum_{k=1}^{N_{\text{ft}}} P_k^{\text{ft}}(t), \quad (8)$$

with  $P^{\text{inf}}$  from Equation (6) at the same share  $w_{\text{main}}$ . The two aggregate classes are then power-matched and both contain the background’s weak lines. Figure 1 shows the four resulting workload classes as a verifier would see them.

### 3. Measurement and threat model

This section specifies how the power drawn by the data centre reaches the verifier (i.e. the measurement model), and what the facility operator can control (i.e. the threat model). The verifier wants to know whether a monitored facility is running undeclared training. We frame this as a hypothesis test. Under the null hypothesis  $\mathcal{H}_0$  the facility runs its declared workload, which for a monitored cluster is dominated by inference: the inference model of Section 2.3, alone or over the background of Equation (8). Under the alternative hypothesis  $\mathcal{H}_1$  a training run of Section 2.1 takes the place of that inference, alone or over the same background (Equation (7)). Section 3.1 outlines the channel through which either hypothesis reaches the verifier. Section 3.2 outlines who controls the load, what they know, and what they may change.

#### 3.1. The measurement model

Section 2 gives the power  $P_{\text{AI}}(t)$  drawn by the AI workload above the facility’s static overhead, in units of  $P_{\text{peak}}$ . The total power behind the meter is

$$P_{\text{fac}}(t) = P_0 + P_{\text{AI}}(t), \quad (9)$$

with  $P_{\text{AI}}$  given by Equation (7) under  $\mathcal{H}_1$ , by Equation (8) under  $\mathcal{H}_0$ , or by a single workload of Section 2 when a result drops the background. The verifier holds one scalar time series from a meter outside the operator’s control. The meter reads

$$P_{\text{meter}}[n] = (h * P_{\text{fac}})(t_n) + \eta[n], \quad (10)$$

$$t_n = n/f_s, \quad h = h_{\text{ins}} * h_{\text{del}}.$$

The meter filters the facility power, samples it at rate  $f_s$ , and adds interference and measurement noise  $\eta[n]$  on its

|                                          | Symbol                                        | Train        | Fine-tune     |
|------------------------------------------|-----------------------------------------------|--------------|---------------|
| Cadence band [Hz]                        | $[f^{\text{lo}}, f^{\text{hi}}]$              | [0.5, 1.5]   | [0.3, 0.7]    |
| Period jitter                            | $\sigma_\xi$                                  | 0.1          | 0.1           |
| Phase ratio                              | $[\rho^{\text{lo}}, \rho^{\text{hi}}]$        | [0.55, 0.8]  | [0.7, 0.9]    |
| Offset spread                            | $\sigma_\Delta$                               | 0.05         | 0.03          |
| Down-phase offset                        | $\mu_\Delta$                                  | 0.3          | 0.8           |
| Up noise range                           | $s^{\text{up}}$                               | [0.02, 0.05] | [0.01, 0.03]  |
| Down noise range                         | $s^{\text{dn}}$                               | [0.01, 0.03] | [0.005, 0.02] |
| Noise scale-down                         | $\kappa_\nu$                                  | 0.15         | 0.15          |
| <i>Cadence drift (dominant run only)</i> |                                               |              |               |
| Reversion [iter <sup>-1</sup> ]          | $\theta$                                      |              | 0.02          |
| Excursion std [Hz]                       | $\sigma_f$                                    |              | 0 (swept)     |
| Drift floor                              | $g_{\text{min}}$                              |              | 0.1           |
| <i>Aggregate</i>                         |                                               |              |               |
| Background counts                        | $N_{\text{tr}}, N_{\text{ft}}$                |              | 4, 4          |
| Power shares                             | $w_{\text{main}}:w_{\text{tr}}:w_{\text{ft}}$ |              | 9 : 0.5 : 0.5 |
| <i>Inference null</i>                    |                                               |              |               |
| Mean serving load                        | $B$                                           |              | 0.5           |
| Decode RMS                               | $A$                                           |              | 0.40          |
| Decode band [Hz]                         | $[f_b^{\text{lo}}, f_b^{\text{hi}}]$          |              | [0.1, 4]      |
| MoE corr. time [s]                       | $\tau_u$                                      |              | 5             |
| MoE depth                                | $m$                                           |              | 0.4           |
| Prefill rate [Hz]                        | $\lambda_p$                                   |              | 0.08          |
| Prefill height                           | $a$                                           |              | 0.15          |
| Prefill duration [s]                     | $d_p$                                         |              | 0.1           |
| <i>Scale</i>                             |                                               |              |               |
| Peak power [W]                           | $P_{\text{peak}}$                             |              | 850           |
| Static overhead [W]                      | $P_0$                                         |              | 250           |

Table 1. Workload-model parameters of Section 2. Training and fine-tuning values are Ko and Zhu’s (Ko & Zhu, 2025), except  $\kappa_\nu$ , the drift parameters, and  $\sigma_\xi$  for fine-tuning (Ko and Zhu list it for training only; we reuse it). The inference null is ours. Ko and Zhu’s intra-phase noise is a sub-millisecond fluctuation, and each sample of our tens-of-hertz generator averages over many of them, so we scale its standard deviation by  $\kappa_\nu$ ; the  $s$  ranges are Ko and Zhu’s values before scaling. Amplitudes are fractions of the peak power  $P_{\text{peak}}$ . The drift excursion  $\sigma_f$  is swept in Section 5. The scale values are ours, and are those of a single accelerator board; only the workload’s size relative to the meter’s additive terms affects detection (Section 3.1).

own sample grid. Round brackets denote continuous time and square brackets index the meter’s samples. The filter is a linear time-invariant system with impulse response  $h$ , which has two parts. The delivery path  $h_{\text{del}}$  is everything between the accelerators and the meter: power supplies, distribution, and any conversion or storage. The instrument  $h_{\text{ins}}$  is the meter’s own filtering: its integration window and calibration gain. Figure 2 traces the power from the workloads to the verifier.

Following Ko & Zhu (2025) we assume the delivery path passes the power through unchanged, such that  $h_{\text{del}}$  is a delta function and  $h_{\text{del}} * P_{\text{fac}} = P_{\text{fac}}$ . The low-pass and notch of Table 2 relax this assumption in our sweeps. The notch stands for an anti-resonance of the delivery network that happens to sit near the cadence; it is the one stage

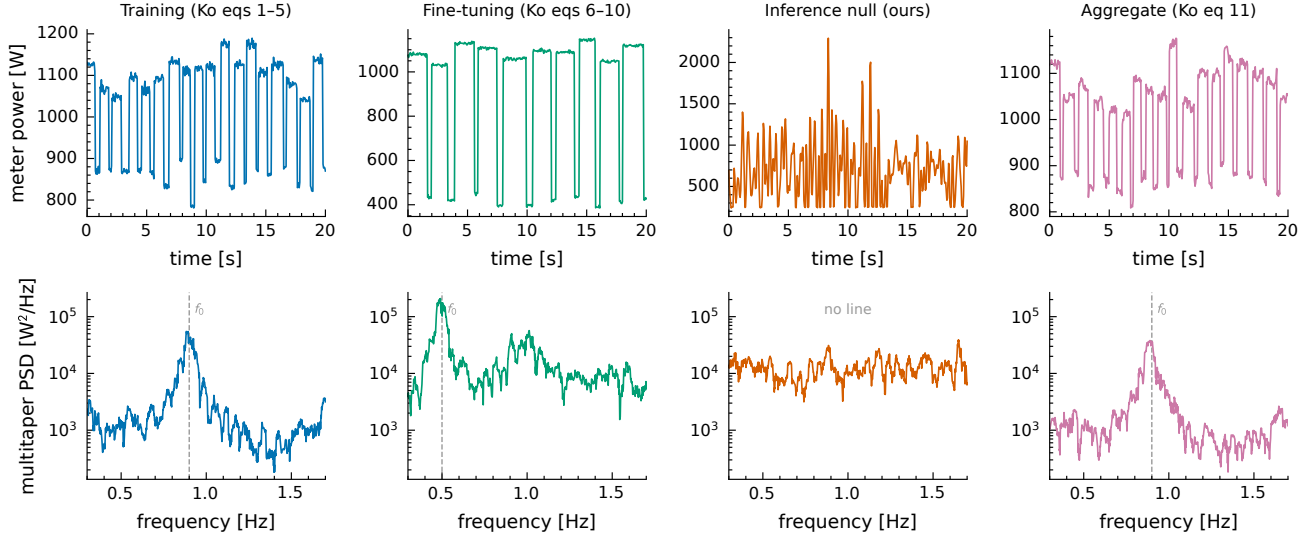


Figure 1. The four workload classes at the power meter, on the nominal channel (20 Hz sampling, no filtering or decimation,  $\sigma_\eta = 4$  W). Top: a 20 s excerpt. Bottom: the multitaper spectrum over the 0.3–1.7 Hz search band, with the true cadence marked. Training, fine-tuning, and the aggregate carry a line at  $f_0$ ; the inference null does not. Cadences are pinned for legibility ( $f_0 = 0.9$  Hz training, 0.5 Hz fine-tuning); elsewhere they are drawn from the Ko bands.

that removes signal in-band while leaving the rest of the spectrum intact.

The instrument  $h_{\text{ins}}$  is the meter’s own response (Table 2). We model it as a calibration gain  $g$  followed by a trailing boxcar, so that each reading is the mean power over the preceding window  $\tau$ :  $h_{\text{ins}}(t) = g/\tau$  for  $0 \leq t < \tau$  and zero otherwise. Its frequency response has magnitude

$$|H_{\text{ins}}(f)| = g \left| \frac{\sin(\pi f \tau)}{\pi f \tau} \right|, \quad (11)$$

which passes slow variations and attenuates fast ones, with a zero at every multiple of  $1/\tau$ . A window longer than about 0.6 s puts the first zero inside the 0.3–1.7 Hz search band, and a 1 s window puts it at the band centre. In the nominal channel  $g = 1$  and  $\tau \rightarrow 0$ , so  $h_{\text{ins}}$  is also a delta function, and the meter samples at  $f_s = 20$  Hz. Unlike the delivery path, the instrument is the verifier’s to choose, and Section 5.2 asks what  $\tau$  and  $f_s$  must be for detection to hold.

The additive term  $\eta[n]$  collects everything that reaches the readings without passing through  $h$  (Table 2). In the nominal channel it is white meter noise of  $\sigma_\eta = 4$  W. Two further terms are available: a slowly wandering baseline, standing for a drifting operating point such as cooling load, and a controller limit cycle, a square wave whose frequency itself wanders, modelled on the power-management oscillation measured on an A100 (Section 7.2). The controller is the first appearance of an instrument line: a persistent, slowly wandering spectral line that the facility’s hardware produces with no training present. Section 7.1 returns to it.

Three numbers carry units. The peak power  $P_{\text{peak}} = 850$  W

and overhead  $P_0 = 250$  W (Table 1) are those of a single accelerator board, and the meter noise  $\sigma_\eta = 4$  W is an illustrative choice, about 0.5% of the mean reading, not a measured value. None of the three is critical.  $P_0$  is constant, so it lies at zero frequency, outside the search band. What sets detectability is the training line against everything else in the band, and at the nominal setting that is dominated by the workloads themselves: in the search band the inference null’s power spectral density exceeds that of the meter noise by about four orders of magnitude, and the training trace’s own off-line floor exceeds it by about three. Meter noise binds only when it is raised far above 4 W, which Figure 3 does over two decades.

Each trace lasts  $T = 300$  s, a frequency resolution of  $1/T \approx 3$  mHz. In the nominal channel every stage of Table 2 is off except the white meter noise. With the noise also off, the map returns  $P_{\text{fac}}$  unchanged and consumes no randomness. Two limits apply. The linear time-invariant form of  $h$  is a first approximation; a real delivery path may be operating-point dependent. And the model quantises only in time, through the integration window and the sample grid; it has no amplitude quantisation. Until simultaneous on-device and external-meter measurement is performed, claims about what reaches an off-chip sensor are predictions (Section 7.2).

Figure 3 shows how the channel attenuates the training line. For a fixed training workload we vary one axis of Table 2 at a time and measure an effective signal-to-noise ratio at the cadence: the spectral power at  $f_0$  over the in-band power the meter itself adds, relative to the nominal channel. Every axis

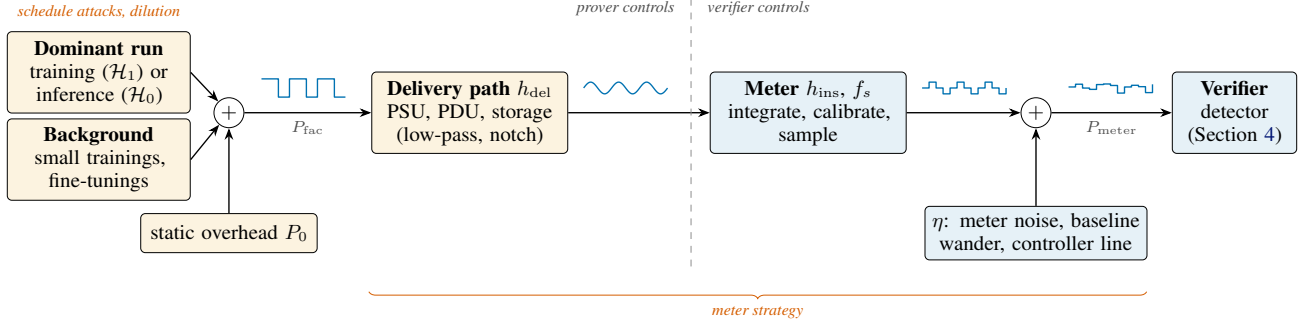


Figure 2. The measurement chain. The workloads of Section 2 and a static overhead sum to the facility power  $P_{\text{fac}}$  (Equation (9)), which passes through the delivery path and the meter before additive terms are added (Equation (10)). Sketches show one training cadence as it is smoothed, sampled, and made noisy; they are illustrative. Everything left of the dashed line is under the operator’s control (Section 3.2). Red labels mark where the adversarial strategies of Table 3 act.  $\eta$  lumps instrument noise with interference that physically originates behind the meter, such as the power-controller line. In the nominal channel the delivery path passes the power through unchanged ( $h_{\text{del}} = \delta$ ) and the meter samples at 20 Hz with no integration (Table 2).

| Stage                                                                                  | Parameter                          | Nominal  | Swept               | Stands for                   |
|----------------------------------------------------------------------------------------|------------------------------------|----------|---------------------|------------------------------|
| <i>Delivery path (<math>h_{\text{del}}</math>; the identity in the workload model)</i> |                                    |          |                     |                              |
| Butterworth low-pass (order 2)                                                         | cutoff [Hz]                        | off      | 8–0.5               | PSU/PDU bandwidth, smoothing |
| In-band notch                                                                          | depth; centre [Hz]; $Q$            | off      | 0.5–1; 0.6–1.2; 1.5 | delivery-path anti-resonance |
| <i>Instrument (<math>h_{\text{ins}}</math> and sampling; chosen by the verifier)</i>   |                                    |          |                     |                              |
| Trailing boxcar integration                                                            | window $\tau$ [s]                  | 0        | 0.1–2               | meter averaging              |
| Gain                                                                                   | $g$                                | 1        | —                   | calibration factor           |
| Sampling (no anti-alias filter)                                                        | $f_s$ [Hz]                         | 20       | 10–0.5              | meter reporting rate         |
| <i>Additive terms (<math>\eta</math>)</i>                                              |                                    |          |                     |                              |
| Meter noise (white or AR(1))                                                           | $\sigma_\eta$ [W]; $\tau_\eta$ [s] | 4; white | 1–120; 2            | instrument noise             |
| Baseline wander (OU)                                                                   | std [W]; corr. time [s]            | off      | 20; 60              | drifting operating point     |
| Controller limit cycle                                                                 | amplitude [W pk-pk]                | off      | 10–120 at 0.38 Hz   | power-management loop        |

Table 2. Stages of the observation map (Equation (10)). Nominal values define the channel used everywhere unless stated. The swept ranges are those of Figure 3 and the minimum-meter sweep of Section 5.2; the hostile channels of the meter strategy (Table 3) are drawn from the same ranges. Every stage off except  $\eta$  is the nominal channel; every stage off including  $\eta$  is the identity.

lowers it, and two remove the line entirely within the range a real meter might occupy: a 1 s integration window, whose first zero falls at 1 Hz (Equation (11)), and a 1 Hz sample rate. This concerns the signal reaching the verifier, not any detector; what each detector does with what survives, and hence the minimum meter specification, is Section 5.2.

### 3.2. The threat model

The verifier installs and owns the power meter. The meter is tamper-evident, the verifier chooses its specification, and the operator cannot forge the meter noise  $\eta$ . The operator controls everything behind the meter, e.g. which jobs run, how they are scheduled, what else shares the facility, and the power-delivery path itself. Energy storage or power smoothing between the accelerators and the meter can low-pass or notch the load, and datacentre and grid operators already treat smoothing training load as an engineering problem (Ko & Zhu, 2025). We represent partial smoothing by the filtering stages of Table 2, and Section 5.2 asks how much of it the verifier can tolerate. A store large enough to flat-

ten the load entirely would erase any passive signature; we do not model it, and discuss it among the limitations (Section 7.4). The operator knows the detector, its parameters, its threshold, and the meter specification. It does not know the realisation of  $\eta$ .

We consider three types of facility operator. First a *compliant* operator runs only its declared inference. A *benign* operator trains without regard to being caught by the power meter. A *concealing* operator trains, knows the detector, and reshapes its workload to try to evade it. The concealing operator’s objective is to train without the being detected by the verifier. They use different adversarial strategies to try to subvert the verifier’s detection. Table 3 lists the adversarial strategies considered in this work. They fall into five categories: disturbing the timing of iterations, relocating the cadence, reshaping the power waveform within an iteration, mixing the run with other load, and degrading the channel. In principle the strategies may be composed, but we treat them individually. Adversarial strategies have a cost to the operator in terms of training throughput. Each

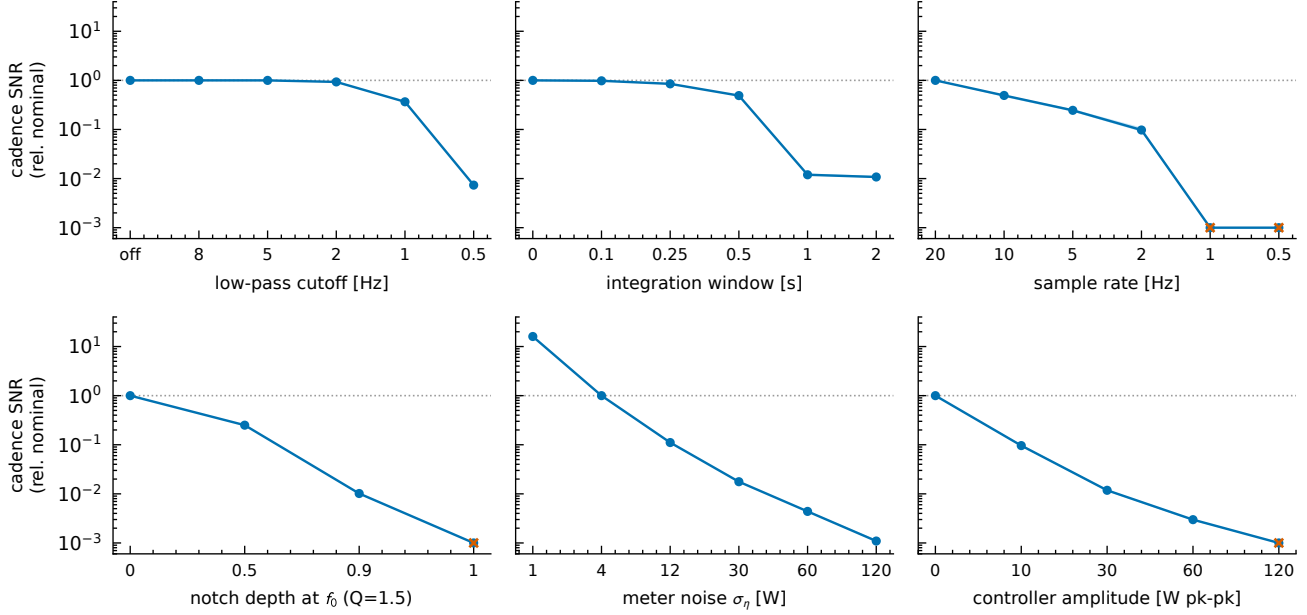


Figure 3. Sensitivity of the cadence signature to the observation map, one axis at a time on a fixed honest training workload ( $f_0 = 0.9$  Hz, 20 realisations, median and interquartile range). The measure is spectral power at the known  $f_0$  over the in-band power the meter itself contributes, normalised to the nominal channel, so that attenuating axes act on the numerator and additive ones on the denominator and every panel reads the same way: down is worse for the verifier. Crosses mark settings where the cadence is not observable at all. Note  $f_0$  is pinned away from 1.0 Hz, which would coincide with both the 2 Hz Nyquist frequency and the first sinc null of a 1 s boxcar.

strategy therefore trades concealment against cost.

Three things lie outside this threat model. First, the operator does not tamper with the meter or forge its noise. Second it does not run a semantic decoy, that is a load that reproduces the physical schedule of training without training, since manufacturing evidence of the activity it is hiding buys a concealing operator nothing. Finally, it does not move the training cadence outside the searched band or abandon synchronous iteration altogether; both are scope restrictions on the population we target (Section 7.4).

## 4. Detection method

In this section we present a detection method that ingests  $P_{\text{meter}}[n]$ , Equation (10), and decides between  $\mathcal{H}_0$  and  $\mathcal{H}_1$  (Section 3). The two hypotheses are separated by the iteration cadence of synchronous training. Under  $\mathcal{H}_1$  the two-phase cycle of the dominant run (Equation (2)) puts a strong, narrow spectral line at its cadence  $f_0$ . Under  $\mathcal{H}_0$  the inference that replaces it spreads its power across the band with no line (Section 2.3), and the only lines left are the background’s weak ones (Section 2.4). A line at a fixed frequency has classical tests. But the cadence of a real run wanders by  $\sigma_f$  even without an adversary (Equation (3)), and a concealing operator can make it wander further or vary the work per iteration, which defeats a fixed-frequency test cheaply (see Section 6). The detector therefore has to

follow the line as it moves.

This section is organised as follows. Section 4.1 defines the detector, a Viterbi line tracker borrowed from passive sonar and continuous-gravitational-wave searches. Section 4.2 sets the threshold at which the detector declares training. Section 4.3 defines the two fixed-frequency tests we compare it against.

### 4.1. Viterbi line tracking

We follow the line with the hidden-Markov frequency tracking developed for passive sonar (Streit & Barrett, 1990) and carried into continuous-gravitational-wave searches (Suvorova et al., 2016; Bayley et al., 2019; Djurović & Stanković, 2011). The tracker has three parts: a spectrogram of the trace, a contrast map built from it, and a search for the best slowly varying path through that map.

Let  $x[n] = P_{\text{meter}}[n] - \bar{P}_{\text{meter}}$  be the mean-removed meter trace,  $N = f_s T$  samples long. We cut it into frames of  $M$  samples that overlap by half, a hop of  $H = M/2$ , and take the power spectrum of each:

$$S_{kj} = \left| \sum_{m=0}^{M-1} w[m] x[jH + m] e^{-2\pi i k m / M} \right|^2, \quad (12)$$

for frames  $j = 0, \dots, N_t - 1$ , where  $w$  is a tapered (Tukey) window and  $N_t = \lfloor (N - M)/H \rfloor + 1$  is the number of frames. Bin  $k$  sits at frequency  $f_k = k f_s / M$ . We choose

| Strategy           | What the operator changes                                                                                             | Budget knob (range)                  | Cost anchor |
|--------------------|-----------------------------------------------------------------------------------------------------------------------|--------------------------------------|-------------|
| <i>Timing</i>      |                                                                                                                       |                                      |             |
| Period jitter      | i.i.d. period jitter $\sigma_\varepsilon$ of Equation (1), by inserting idle time                                     | $\sigma_\varepsilon \in [0.05, 0.7]$ | measured    |
| Cadence drift      | OU excursion $\sigma_f$ of Equation (3)                                                                               | $\sigma_f \in [0, 1.5]$ Hz           | measured    |
| Work variation     | micro-steps per iteration $G_i$ , drawn about $G_0 = 8$ with fractional spread $\sigma_G$ ; down-phase unchanged      | $\sigma_G \in [0.05, 0.7]$           | measured    |
| Phase slip         | Brownian slip of iteration boundaries at fixed mean period                                                            | slip std $\in [0.01, 0.2]$ periods   | —           |
| <i>Frequency</i>   |                                                                                                                       |                                      |             |
| Relocation         | cadence $f_0$ moved toward the search-band edges                                                                      | $f_0 \in [0.35, 1.65]$ Hz            | —           |
| <i>Waveform</i>    |                                                                                                                       |                                      |             |
| Harmonic smoothing | raised-cosine ramp on phase transitions; comb dies, fundamental survives                                              | ramp width $\in [0.1, 1]$ s          | —           |
| Shape filling      | down-phase level raised toward the up-phase at matched mean power                                                     | fill fraction $\in [0, 1]$           | measured    |
| <i>Mixing</i>      |                                                                                                                       |                                      |             |
| Dilution           | training’s share of the aggregate (Equation (7)) lowered                                                              | share $\in [0.2, 0.9]$               | —           |
| <i>Channel</i>     |                                                                                                                       |                                      |             |
| Meter              | hostile $h$ and $\eta$ : notch at the cadence; 1 s/1 Hz integrating sampler; in-band controller; heavy coloured noise | four named channels                  | —           |

Table 3. Adversarial strategies available to the concealing operator. Each is implemented inside the generator of Section 2 and swept singly over the stated budget range in Section 6. “Measured” marks strategies with a throughput-overhead anchor from the single-A100 campaign (Section 7.2); the others have no hardware implementation and are reported without a cost (Section 6). The meter strategy degrades the channel rather than the schedule and is read as a channel requirement (Section 5.2), not priced.

$M$  so that each frame lasts  $M/f_s = 16$  s, which gives bins  $\Delta f = 1/16 = 62.5$  mHz wide, fine enough to resolve a line near 1 Hz, while leaving many frames to track over: a 300 s trace gives  $N_t = 36$  frames, 8 s apart. Only the bins with  $f_k \in \mathcal{B}$  enter the tracker, 23 of them in the nominal channel. Every detector in this paper searches the band  $\mathcal{B} = [0.3, 1.7]$  Hz. This is the training cadence band  $[f^{\text{lo}}, f^{\text{hi}}] = [0.5, 1.5]$  Hz of Table 1, widened by 0.2 Hz on each side so that a cadence drawn near either edge stays in the band as it wanders.

We normalise each frame by its median over the band,

$$L_{kj} = \ln S_{kj} - \ln \tilde{S}_j, \quad (13)$$

where  $\tilde{S}_j$  is the median of  $S_{kj}$  over the in-band bins of frame  $j$ .  $L_{kj}$  measures how far bin  $k$  stands above the typical in-band level of its own frame. The normalisation keeps a slow change in overall power from masquerading as a line.

The hidden state is the frequency bin occupied by the cadence in each frame. The prior that a real cadence wanders slowly enters as a penalty  $\lambda$  per bin of frequency change between consecutive frames. The detection statistic is the score of the best path, averaged over frames,

$$V = \frac{1}{N_t} \max_{k_0, \dots, k_{N_t-1}} \left[ \sum_{j=0}^{N_t-1} L_{k_j j} - \lambda \sum_{j=1}^{N_t-1} |k_j - k_{j-1}| \right], \quad (14)$$

with  $\lambda = 1$  throughout. A path pays one unit of log-contrast for each 62.5 mHz step it takes between frames, so it can

follow a drift of about one bin per frame (about 8 mHz s<sup>−1</sup>) if the line it follows is bright enough to pay for it. Dynamic programming evaluates the maximum over all  $23^{N_t}$  paths exactly, in time linear in  $N_t$ . A drifting line keeps a high score because the path bends to follow it. Broadband power, such as the continuous-batching floor of the inference null, does not, because no single slowly varying path stays on top of it. Figure 4 shows both cases on the contrast map. Averaging over frames makes  $V$  comparable across traces of different length. Note that  $V$  tests whether a persistent, slowly wandering line is present, not what produced it. Accordingly, strong lines that do not have a training provenance may also trigger the detector; this is addressed in Section 7.1

## 4.2. Setting the threshold

The tracker declares training when  $V$  exceeds a threshold  $v_\alpha$ , chosen so that the false-alarm rate is  $\alpha$ :

$$\Pr(V > v_\alpha \mid \mathcal{H}_0) = \alpha. \quad (15)$$

We use  $\alpha = 10^{-2}$  unless stated.  $V$  has no analytic null distribution. We therefore set  $v_\alpha$  empirically, as the  $(1 - \alpha)$  quantile of  $V$  over simulated traces drawn under  $\mathcal{H}_0$ , and measure detection rate as the fraction of  $\mathcal{H}_1$  traces above it. The baseline methods of Section 4.3 are thresholded the same way.

In Section 5 the  $\mathcal{H}_0$  population is the null aggregate of Equation (8). Its background carries the weak lines of the small trainings and fine-tunings, so the threshold already accounts for any line that is not the dominant run’s. Calibrating on

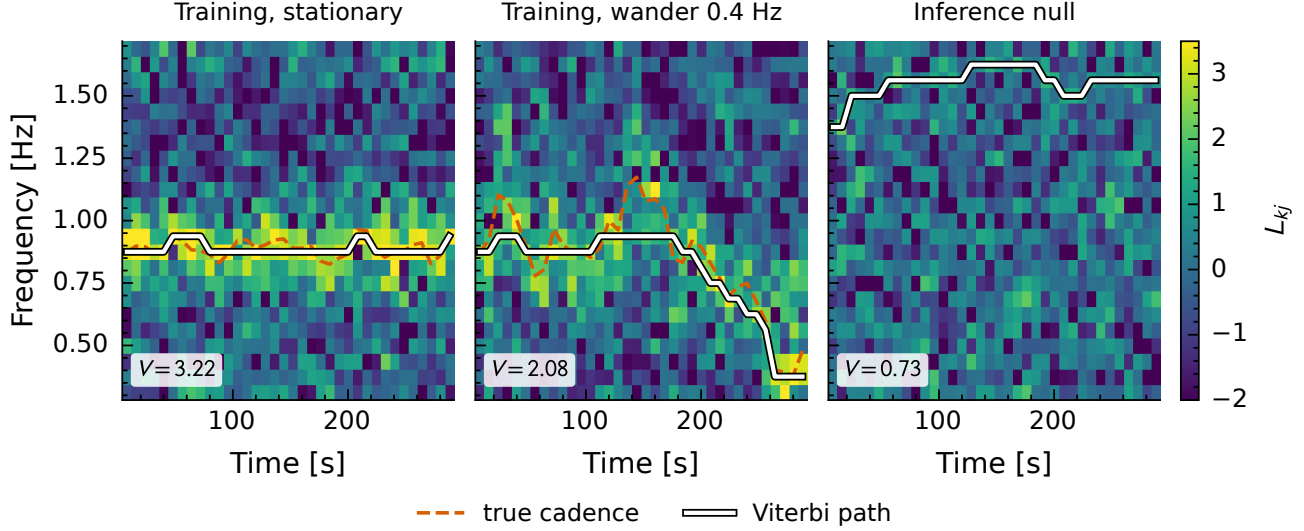


Figure 4. The Viterbi tracker on three 300 s traces at the nominal channel. Each panel is the contrast map  $L_{kj}$  of Equation (13) on the tracker’s own grid (16 s frames at half overlap, 62.5 mHz bins over  $\mathcal{B}$ ), with the best path of Equation (14) in white and its score  $V$  inset. Dashed: the true cadence,  $1/(\text{median iteration period})$  in each frame, from the generator’s iteration boundaries. Left: a stationary training cadence ( $f_0 = 0.9$  Hz), which the path sits on. Centre: the same run with 0.4 Hz of cadence wander, which the path follows across most of the band; it cuts the corner off the brief excursion near 150 s, where following the line costs more in jumps than it gains in contrast. Right: the inference null, where no path finds sustained contrast and  $V$  falls to 0.73. Reproduced by `scripts/plot_viterbi_track.py`.

inference alone, with no background, changes the tracker’s detection rates by at most 0.02 (Appendix C).

This calibrates  $V$  against the modelled inference null. A deployed verifier has to obtain the same threshold from data it holds: from background traces known to be free of training, as continuous-wave searches set their tracking thresholds from noise-only realisations (Suvorova et al., 2016), or from a per-trace surrogate, e.g. Appendix B.

#### 4.3. Baseline method

Two simple fixed-frequency tests serve as baselines to compare our Viterbi method against. The first is a spectral matched filter, which uses the peak of the power spectral density in the band, relative to its median over all frequencies,

$$D_{\text{MF}} = \frac{\max_{f \in \mathcal{B}} \hat{S}(f)}{\text{median}_f \hat{S}(f)}, \quad (16)$$

where  $\hat{S}$  is a Welch estimate of the spectrum of  $x[n]$  from half-overlapping Hann-windowed segments of 1024 samples (51.2 s at  $f_s = 20$  Hz). The long segments concentrate a stationary line into one bin, which is the test’s strength; a wandering line smears across many bins and the peak collapses. The second is the Thomson multitaper harmonic F-test (Thomson, 1982; Percival & Walden, 1993), maximised over the band,

$$D_F = \max_{f \in \mathcal{B}} F(f), \quad (17)$$

where, at any fixed  $f$  under a stationary, locally smooth null,  $F(f) \sim F(2, 2K - 2)$ , with  $K = 7$  tapers of time-bandwidth product  $NW = 4$ .  $F(f)$  is the ratio of the power in a line at  $f$  to the power left around it (Equation (21), Appendix A).

### 5. Detection results

In this section we test the Viterbi tracker of Section 4 on synthetic meter traces. We assess if the Viterbi tracker can detect the training frequency, and what sampling settings are required by the power meter for the Viterbi tracker to work. We treat the operator as benign; an operator who reshapes the schedule to hide the line is the subject of Section 6. Under  $\mathcal{H}_1$  the meter sees a dominant training run over  $N_{\text{tr}} = 4$  small trainings and  $N_{\text{ft}} = 4$  fine-tunings (Equation (7)). Under  $\mathcal{H}_0$  it sees the null aggregate, with inference in the dominant slot over the same kind of background (Equation (8)). The shares are the defaults of Table 1, 9 : 0.5 : 0.5. Both classes therefore also carry the background’s weak lines, and the tracker has to find the dominant one among them.

This section is organised as follows. Section 5.1 measures the tracker’s detection rate as the cadence drifts, and compares it with the fixed-frequency tests of Section 4.3 and with an optimality ceiling. Section 5.2 sweeps the meter and states the minimum specification under which detection holds.

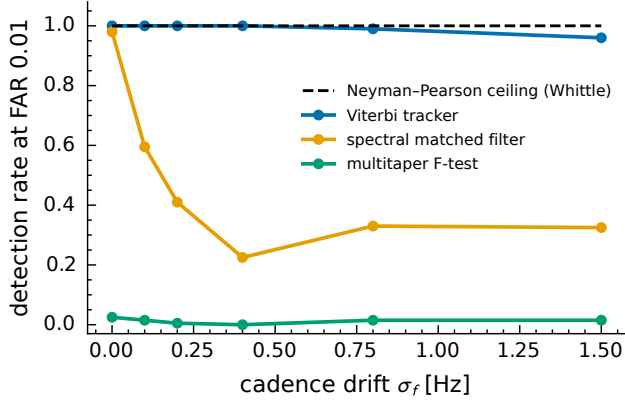


Figure 5. Detection rate at empirical false-alarm rate  $10^{-2}$  against the null aggregate, versus cadence drift  $\sigma_f$  of the dominant training run ( $n = 200$  per class, 300 s traces, shares 9 : 0.5 : 0.5). The Viterbi tracker stays within 0.04 of the Whittle Neyman–Pearson ceiling across the wander axis; the spectral matched filter collapses once the line moves and the multitaper F-test has no usable power. The ceiling is described in Appendix C.

### 5.1. Detection rate under cadence wander

We consider a range of cadence-drift  $\sigma_f \in \{0, 0.1, 0.2, 0.4, 0.8, 1.5\}$  Hz (Equation (5)) in the dominant training run, with  $n = 200$  traces of 300 s per class and thresholds set at an empirical false-alarm rate of  $10^{-2}$ . We benchmark the Viterbi method against a Neyman–Pearson optimal ceiling (Appendix C). Figure 5 shows the result. The Viterbi tracker stays within 0.04 of the optimal ceiling across the whole  $\sigma_f$  axis, with detection rate 1.00 up to 0.4 Hz of drift, 0.99 at 0.8 Hz and 0.96 at 1.5 Hz. Conversely, the baselines fixed tests struggle. The spectral matched filter detects a stationary line (0.98 at zero drift) but falls to 0.59 at 0.1 Hz and 0.23 at 0.4 Hz, and the multitaper F-test has essentially no power at empirical thresholds at any drift ( $\leq 0.03$ ).

### 5.2. A minimum meter specification

The results of Section 5.1 assume the nominal meter of Table 2: a sample rate of 20 Hz, no integration window ( $\tau \rightarrow 0$ ), no filtering on the delivery path, and white meter noise of  $\sigma_\eta = 4$  W. A real meter may sample more slowly, average over a longer window, or sit behind a delivery path that suppresses the cadence band. The meter is the verifier’s choice, so it is of interest to find the minimum specification it must meet for detection to hold. We sweep the sample rate over  $f_s \in \{20, 10, 5, 2, 1, 0.5\}$  Hz and the integration window over  $\tau \in \{0, 0.1, 0.25, 0.5, 1, 2\}$  s (Equation (11)). We also sweep a notch in the delivery path, i.e. a band near the cadence in which the delivery path suppresses the power swing (Section 3.1). Its centre ranges over 0.6–1.2 Hz and its depth, the fraction of the signal it removes at the centre, over 50–100%, with the meter otherwise nominal.

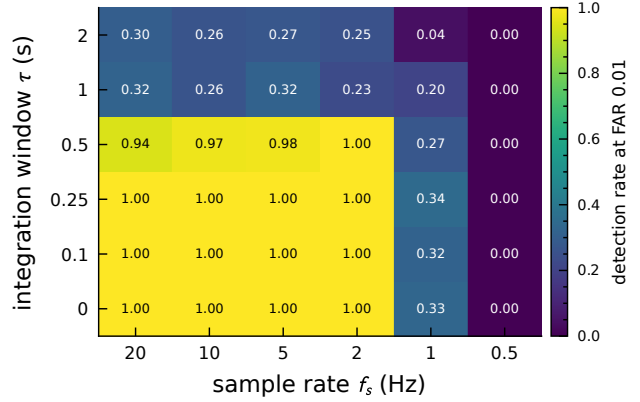


Figure 6. Minimum meter specification: Viterbi detection rate at empirical false-alarm rate  $10^{-2}$  over sample rate  $f_s \times$  integration window  $\tau$ , the honest aggregate against the null aggregate ( $n = 200$  per class). The live region is  $f_s \geq 2$  Hz and  $\tau \leq 0.5$  s.

Throughout, the honest aggregate is scored against the null aggregate, with thresholds at an empirical false-alarm rate of  $10^{-2}$  as in Section 5.1.

The Viterbi tracker holds full power (detection rate  $\gtrsim 0.9$ ) if and only if the meter satisfies three conditions together (Figure 6), one on each part of Equation (10). First, on the sampling, a rate of at least about 2 Hz: rates of 20, 10, 5, and 2 Hz all hold the Viterbi tracker near 1.0, 1 Hz collapses it to 0.33 as the Nyquist edge crosses the band, and 0.5 Hz is dead (every detector at chance). Second, on the instrument’s impulse response, an integration window  $\tau \lesssim 0.5$  s: at 20 Hz sampling  $\tau = 0.5$  s still holds (0.94), but  $\tau = 1$  s puts the first zero of Equation (11) at  $1/\tau = 1$  Hz, the band centre, and collapses the tracker to 0.32 even at full sample rate. Third, on the delivery path, no deep transfer-function notch at the cadence: a shallow notch, or one near the edge of the cadence band, costs little (tracker  $\geq 0.80$ ), but a notch of 90–100% depth at 1.0–1.2 Hz roughly halves tracker power (to 0.45–0.63) without erasing it (Appendix E).

Against an honest operator, then, detection needs only a meter that samples at  $f_s \geq 2$  Hz, integrates over  $\tau \leq 0.5$  s, and has no deep notch at the cadence.

## 6. Detection against a concealing operator

Section 5 assumed a benign operator, who leaves the cadence for the tracker to find. A concealing operator can instead de-periodicise its schedule and smear the cadence away (Section 3.2). In principle this is always possible: the concealing operator can insert idle time at random between iterations, or vary the work done in each, to remove any frequency for the Viterbi detector to identify. The question is the cost in training throughput the operator must incur in order to de-periodicise enough to hide from the Viterbi detector, and in particular whether any de-periodicisation

strategy is both cheap and effective.

In this section we sweep all eight schedule strategies of Table 3 and measure how much each one hides from the detectors at each strength. The operator applies each strategy to its dominant training run,  $P^{\text{main}}$  in the aggregate of Equation (7). The background of small trainings and fine-tunings,  $P_j^{\text{tr}}$  and  $P_k^{\text{ft}}$ , stays honest, and every cell is scored against the aggregate null of Equation (8). For each strategy we step its budget knob over the range in Table 3. We measure the detection rate of the Viterbi tracker and of the fixed tests of Section 4.3, taking the better of the two fixed tests at each cell. All rates are at false-alarm rate  $10^{-2}$  with  $n = 200$  traces per class.

this para to complete once we get results back on other ways to measure cost Four strategies — period jitter, cadence drift, work variation and shape filling — were also run on a single A100, which gives each of their knob settings a measured throughput cost (Section 7.2). We write the throughput cost of a setting as

$$c = \frac{r_{\text{honest}}}{r} - 1, \quad (18)$$

the fractional increase in wall-clock time per trained token, where  $r$  is the training throughput in tokens per second under the attack and  $r_{\text{honest}}$  that of the same run without it, and quote it in percent. For the strategies that only insert idle time,  $c$  is the stretch of the iteration period over its busy time; for work variation, which never idles, it is read from the tokens-per-second deficit  $1 - r/r_{\text{honest}}$ , which agrees with Equation (18) to first order and is zero within noise. The other four (phase slip, relocation, harmonic smoothing and dilution) have no hardware implementation, so they have no measured cost. Throughput is the only cost we measure. An attack might keep tokens per second but worsen progress per token; this learning-efficiency penalty cannot be measured in a synthetic model and remains open (Section 7.4). The full sweep and its provenance are in Appendix D.

This section is organised as follows. Section 6.1 shows how much each strategy hides as its budget knob is turned up, against both detectors; it is the section’s main result. ?? explains why varying the real work per iteration is the one strategy that hides for free, and why at its largest budget it also halves the tracker’s detection rate.

### 6.1. How much each strategy hides

Figure 7 plots hiding ( $1 - \text{detection rate}$ ) against the budget knob of every schedule strategy, with the measured throughput cost of each knob setting where one exists.

Against the fixed tests, seven of the eight strategies hide a large fraction of runs at their larger budgets; only harmonic smoothing hides nothing. Hiding mostly grows with the budget, up to 0.88 under both jitter and work variation. Only work variation hides for free. It hides 0.88 at  $c \approx 0$ ,

while jitter and drift need  $c = 159\%$  and  $356\%$  to reach 0.88 and 0.71.

Against the Viterbi tracker almost nothing hides. The tracker detects at least 95% of runs in 43 of the 46 cells. It detects every run in every cell of phase slip, relocation, harmonic smoothing, shape filling and dilution, and at least 97% under drift. The three exceptions are two jitter settings and the largest work variation. Jitter at  $\sigma_\xi = 0.35$  hides 0.24 for  $c = 159\%$ , and at 0.5 hides 0.12 for  $c = 375\%$ ; at 0.7 it hides nothing. Work variation at  $\sigma_G = 0.7$  hides 0.55 at  $c \approx 0$ , though at every smaller  $\sigma_G$  it hides at most 0.05.

The major challenge for the verifier and the Viterbi tracker is work variation at  $\sigma_G = 0.7$ .

Here a few iterations accumulate very many micro-steps: about one in ten accumulates 64 or more, against the honest  $G_0 = 8$ . These long, unbroken compute blocks carry the hiding; capping the micro-step count at 32 restores the tracker’s detection rate to 84%. Because every micro-step is real work and the device never idles, the throughput cost  $c$  is zero within noise at every  $\sigma_G$  (Figure 7).

The attack is not free in learning efficiency, however. The same long blocks raise the mean micro-step count from 8 to 25, so the operator takes about a third as many optimiser steps per token as honest training. Larger batches make each step less efficient per token. McCandlish et al. (2018) model an optimiser step on a batch of  $B$  samples, with the learning rate tuned for that batch, as making a fraction

$$h(B) = \frac{B}{B + B_{\text{noise}}} \quad (19)$$

of the progress of a full-gradient step, where the gradient noise scale  $B_{\text{noise}}$  sets the critical batch size. Progress per sample is then proportional to  $h(B)/B = 1/(B + B_{\text{noise}})$ , which falls as the batch grows. For micro-step counts  $G_i$  drawn independently, it is proportional to  $\mathbb{E}[h(G)]/\mathbb{E}[G]$ , so the attack needs a fraction

$$\Delta = \frac{h(G_0)/G_0}{\mathbb{E}[h(G)]/\mathbb{E}[G]} - 1 \quad (20)$$

more samples than honest training to reach the same loss, with  $B$  and  $B_{\text{noise}}$  counted in micro-batches. For  $B_{\text{noise}}$  between 8 and 128 micro-batches, Equation (20) gives  $\Delta = 53\text{--}184\%$  at  $\sigma_G = 0.7$ , the larger penalty at the smaller noise scale. The attack moves its cost from throughput into learning efficiency rather than avoiding it.

These learning costs are model predictions, not measurements. They assume a learning rate retuned at every step, and  $B_{\text{noise}}$  for a given training run is unknown. Measuring them requires training to a fixed loss under each schedule, which we leave open (Section 7.4). A longer observation window may also recover more of the spread power for

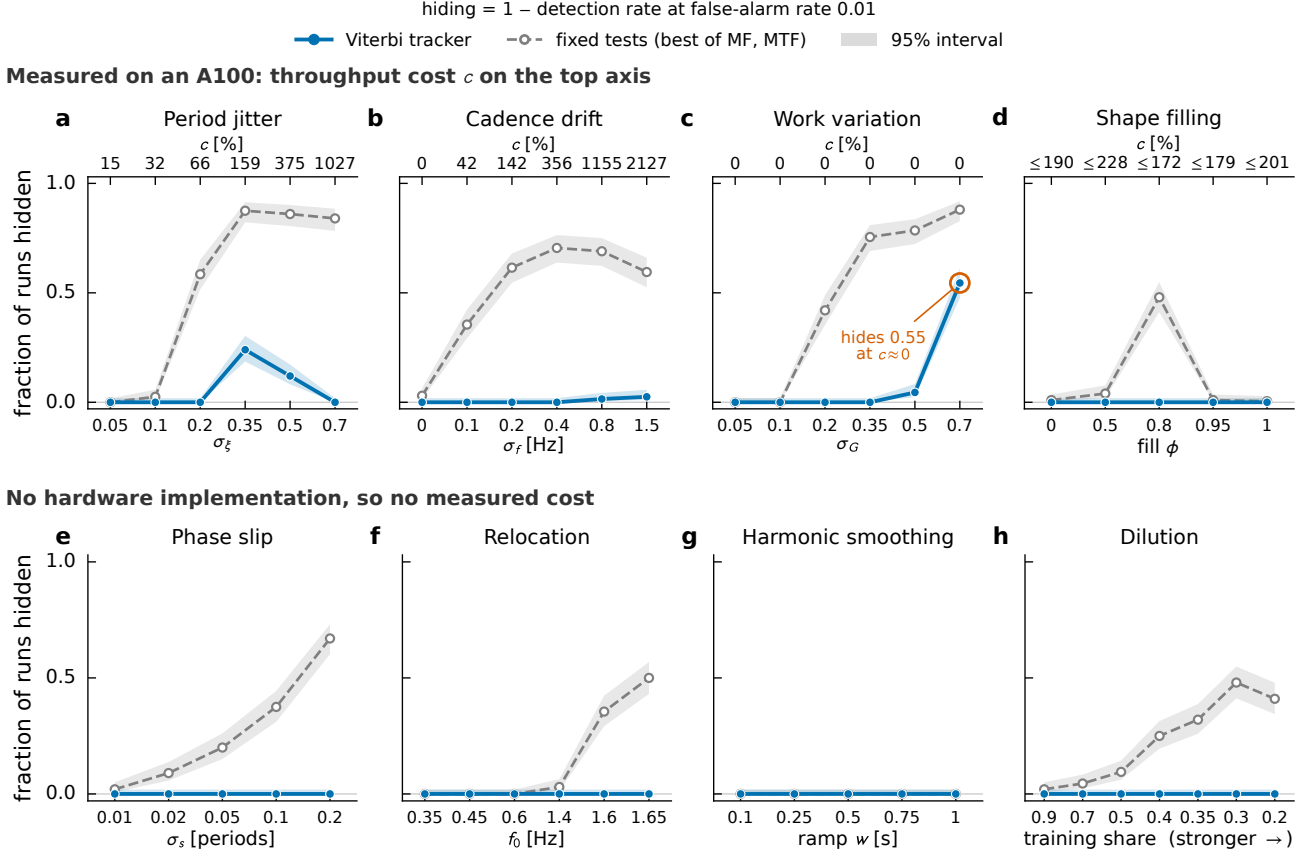


Figure 7. How much each schedule strategy hides, and what it costs. Each panel steps one strategy of Table 3 through its budget knob (bottom axis; settings evenly spaced, weakest on the left) and plots hiding,  $1 - \text{detection rate at false-alarm rate } 10^{-2}$ , for the Viterbi tracker (blue, solid) and the fixed tests (grey, dashed; the better of the matched filter and multitaper F-test per cell). Bands are Wilson 95% intervals for  $n = 200$  traces per class. **a–d**, the four strategies run on a single A100. The top axis gives the measured throughput cost  $c$  of each setting (Equation (18)); costs below 1% in magnitude, zero within noise, are shown as 0, and drift-free scheduling costs zero by definition. Shape-filling costs were measured on a jittered base and are upper bounds ( $\leq$ ). **e–h**, the four strategies with no hardware implementation and so no measured cost. The tracker detects almost every cell. The exception, ringed in **c**, is work variation at  $\sigma_G = 0.7$ , which hides 0.55 from it at  $c \approx 0$ ; idle-insertion jitter hides at most 0.24, for  $c = 159\%$ . The meter strategy degrades the channel rather than the schedule and is treated in Section 5.2. Reproduced by `scripts/plot_attack_curves.py`.

the tracker. [TODO: the work-variation numbers in the last three paragraphs come from the exploratory E0 spike (the E0 findings note under notes/results on the frontier-aggregate-explore branch); promote it to a script with a committed results file before submission. Only the Viterbi tracker was run.]

## 7. Discussion and conclusion

We set out to build a method that turns a low-rate external power trace into a test for iteration-structured training, to price its evasion against an operator who knows the test, and to state what the channel cannot certify at any price. The method is a Viterbi line tracker. Against the inference null that stands for a facility’s declared workload it is essentially Neyman–Pearson optimal across cadence wander (Section 5.1), provided the meter samples at  $\geq 2$  Hz with

$\leq 0.5$  s integration (Section 5.2). Varying real work per iteration evades every fixed-frequency test at no measurable throughput cost. The tracker survives it until the variation is extreme, where it still detects about half of runs; that extreme level is also free in throughput, and every attack that does cost throughput hides less from the tracker (Section 6). This section says what such a detection certifies, where the synthetic evidence stops, and how the result reads for a regulator.

### 7.1. What a detection certifies

A detection establishes that the trace carries a training-shaped periodic schedule. It does not establish that the computation is training. We built seven control loads whose training status we know and scored them with three decision rules fitted to separate training from the inference null

(Appendix F). Five non-training loads — a training-shaped non-ML kernel loop, a controller limit cycle, a periodic inference request generator, a gradient-only computation, and a discarded-update decoy that runs the full forward, backward and optimiser step and throws the update away — all score as training under every rule (Table 4).

The adversary operators of Section 3.2 deliberately omit the extremal member of this class. A *semantic decoy* — a load reproducing the physical schedule of training without retaining a trained model — is not an attack a concealing operator would mount, since manufacturing evidence of the activity it is hiding buys it nothing. It enters this paper instead as a non-training control (Table 4) and as the construction behind the non-identifiability argument below, and it carries weight for two reasons that need no motivated decoy-runner: its benign counterparts occur in ordinary datacentre operation, and the active protocol of Appendix H inverts the operator’s objective, at which point a *challenge-aware decoy* — a correlated dummy response mimicking the prescribed reaction without training — does become a live attack.

The ceiling is a property of the channel, not of our classifiers. Write  $w$  for a computational execution,  $h$  for the hardware and power-delivery path, and  $K_h(w)$  for the law of the observed trace that Equation (10) induces. Call two executions *observationally equivalent* when  $K_{h_0}(w_0) = K_{h_1}(w_1)$ . A passive verifier sees only the trace, so its decision is some measurable function of it, and on an observationally equivalent pair every such function fires with the same probability under both. No passive scalar-power detector can therefore certify the semantic content of a computation against an unrestricted adversary. A non-vacuous guarantee has to put the equivalent pairs out of scope, and there are four ways to do that within the passive channel — restricting the admissible non-training workload class, the adversary’s freedom to add or restructure computation, the hardware and power-transfer family, or the physical property claimed — alongside a fifth route that leaves the passive channel altogether by changing the protocol (Appendix H). We do not claim the list is exhaustive.

The construction behind this is nearly available on the operator’s own hardware. A program can execute the same forward, backward, and optimiser kernels as training and discard each updated state, making its trace arbitrarily close to genuine training while its semantic classification differs. This is not a hypothetical weakness; it is what Table 4 measures. That outcome is a property of the channel rather than of our classifiers, which is why this ceiling cannot be removed by a better detector and why escaping it requires an active challenge (Appendix H).

The argument binds even where no adversary is present. A concealing operator has no reason to run a decoy (Section 3.2), but most of the equivalence class is not adversarial

at all: a training-shaped non-ML kernel loop, a controller limit cycle, and a periodic inference request generator all sit inside it, and all arise in ordinary operation. The consequence is two-sided. A rule phrased semantically — training rather than serving, which model, whose weights — cannot be adjudicated from the schedule alone in either direction, so the same equivalence that would let a decoy counterfeit training also caps the evidentiary weight a verifier can attach to a positive detection.

**Instrument lines and false alarms.** The same reading covers the tracker’s one systematic failure. A periodic physical confound, such as the power-controller oscillation measured on an A100 (Section 7.2), is a persistent wandering line, and the Viterbi score cannot tell it from training (Section B.4). The failure matters for honest operators rather than concealing ones: a controller line does not help an operator hide training, but it can flag a compliant inference facility. Three things bound it. First, a known-line veto. Continuous-wave searches meet the same problem in instrumental lines, and handle it by cataloguing lines from detector characterisation data and discarding candidates coincident with them. The analogue here is to extract several candidate paths from the contrast map (Equation (13)) and discard those that coincide with lines catalogued from the facility’s own characterisation data. We have not evaluated it, and it carries a cost a gravitational-wave veto does not: the adversary knows the veto list, and can relocate its cadence into a vetoed band. Second, per-card controllers need not add coherently at a facility meter (Section 7.2). Third, a positive detection is a trigger for escalation rather than a verdict, and the active protocol of Appendix H resolves a controller false alarm as it resolves the decoys above. The order-tracked cyclostationary statistic of Appendix B partly separates the modelled controller from training by waveform shape, but only conditionally, and we do not rely on it.

## 7.2. Reality check and transfer limits

Every result above is synthetic. The one place we hold the scenario model against measured hardware, it fails in an instructive way, and we report that failure as a result rather than a closing nod.

**The negative transport case.** On measured single-A100 traces the iteration cadence is present but shallow: a saturated card has no communication phase, so the iteration structure modulates a  $\sim 370$  W draw by only a few watts ( $\sim 0.4\%$  of board power). Worse for the naive reading, the spectra contain a 0.31–0.45 Hz component, roughly ten times stronger than the training line, that appears whenever the load varies in time but not at idle or under constant load — the signature of the card’s own power-management

Table 4. Semantic-control outcomes: fraction of each load scored as training at a false-positive rate of 0.05, per decision rule. Five non-training loads all score as training; the only control the tracked rules degrade on is asynchronous (de-periodicised) genuine training, which they nonetheless mostly catch.

| Control load           | Training? | Physics | Fitted | Learned |
|------------------------|-----------|---------|--------|---------|
| Discarded-update decoy | no        | 1.00    | 1.00   | 1.00    |
| Gradient-only          | no        | 1.00    | 1.00   | 1.00    |
| Non-ML kernel loop     | no        | 1.00    | 1.00   | 1.00    |
| Controller cycle       | no        | 1.00    | 1.00   | 1.00    |
| Periodic inference     | no        | 1.00    | 1.00   | 1.00    |
| Co-resident mixture    | yes       | 1.00    | 1.00   | 1.00    |
| Asynchronous training  | yes       | 0.72    | 0.55   | 1.00    |

loop. It is a hardware line that says “time-varying load,” not “training,” and it is the instrument line of Section 7.1: the standing counterexample to reading any detected line as training.

**Why this is the honest outcome.** Two things make the single-card signature shallow. One saturated card has no synchronous communication barrier, so the deep compute-communication contrast the scenario model assumes is absent; and the generator is literature-parameterised, not calibrated to the measured channel, a gap of about  $75\times$  in modulation depth. The synthetic results are therefore a sensitivity analysis over a plausible parameter family, not a prediction calibrated to this hardware — which is why the paper’s claims are scoped to the stated generator and its stated nulls throughout.

**The single-card gap is shared, and lands differently.** The measured side-channel corpus that separates training from inference at high bandwidth is also a single card, with cross-device generalisation left open (Gargiulo & Kulp, 2026). What differs is which observable the gap touches. A board-level probe reads the optimizer-update phase, which is present on one card, so the absent distributed setting costs that result little. The cadence is a collective effect of synchronisation, so a single card is exactly the case in which it is weakest — which is why the same gap is a transport caveat there and a first-order limitation here. The two results also bound each other. The probe’s signature sits below about 20 kHz and survives decimation to about 38 kS/s, three to four orders of magnitude above the few hertz our specification needs, so the fine structure a probe exploits is gone by the time a facility meter reads, and whatever survives at a few hertz must be a slower, more collective observable.

**Measured evasion costs agree.** Work on real hardware prices de-periodicisation the same way as our anchors, through a different channel. Four evasion strategies run on a single H200 — chunked optimizer updates, dilution with decoy inference, throttling, and diluted LoRA — cost between 28 and 69% of throughput (Gargiulo & Kulp, 2026). Each inserts or substitutes work, which is what makes it ex-

pensive, and none varies the real computation per iteration, the one strategy Section 6 finds cheap.

**Falsifiable predictions.** The model makes three predictions we cannot test here but state as falsifiable. Distributed synchronisation should deepen the line: at cluster scale the periodic all-reduce barrier produces a deep, coherent, legible cadence rather than the shallow single-card ripple. And an external wall or rack meter should see a different, lower-passed trace than on-board NVML telemetry, shifting where the minimum meter specification of Section 5.2 binds. And per-card power-controller oscillations, which dominate the single-card spectrum, should add incoherently across a cluster — roughly as  $\sqrt{N}$  for  $N$  independent controllers, against  $N$  for a synchronous training line — and so fall relative to the training line at a facility meter. All three are predictions, not results.

### 7.3. The claim ladder

The results can be read back as a *claim ladder* (Table 5): four claims of increasing governance relevance, each bought at a stated cost in verifier information, and each with a hard ceiling that motivates the next. Read as a procedure, it is screening followed by escalation. Rung 1 establishes that coherent periodic power structure is present, which is evidence rather than identification. Rung 2 converts that structure into training-likeness at the price of committing to an inference null; the detection results of this paper live here, and its ceiling is semantic non-identifiability. Rungs 3 and 4 leave the passive channel for an active protocol and a transcript primitive, and are specified but not run (Appendix H). The ladder is a ladder of claims and assumptions, not a quantitative bound on unidentified compute.

### 7.4. Scope, limitations, and governance reading

**Scope.** Three restrictions fix what the results cover. First, the observable: we use the time-resolved power trace and exclude the integrated-energy channel, so there is no bounding of how much compute was performed and no energy-per-token pricing as a verification instrument. This is a restric-

Table 5. The claim ladder. Each rung answers a strictly more governance-relevant question at a stated cost in verifier information, and each rung’s hard ceiling motivates the next. The fourth column attaches this paper’s evidence.

| Rung                              | Claim                                                                      | Additional requirement                                                | re-        | Evidence in this paper                                                                                                                                                                                                                                                                                      | Hard ceiling                                                                                                                               |
|-----------------------------------|----------------------------------------------------------------------------|-----------------------------------------------------------------------|------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Structural evidence            | Coherent periodic power variation exists under a specified structural null | External resolved only                                                | time-trace | Order-tracked cyclic structure is detected with surrogate-calibrated, level-safe false-alarm control on linear-stationary nulls (Appendix B); the Viterbi score alone cannot exclude a controller (Section B.4)                                                                                             | Periodic physical confounds; structural-null misspecification; die-to-meter attenuation                                                    |
| 2. Conditional classification     | The trace is more training-like than the stated inference population       | A representative or explicitly modelled inference null                |            | Viterbi detects training at the Neyman–Pearson ceiling across cadence wander (Section 5.1) and against zero-cost variable real work (Section 6), given a meter of $\geq 2$ Hz and $\leq 0.5$ s integration (Section 5.2); fitted rules reach area under the curve 1.00 under six domain shifts (Appendix F) | Domain shift; strong superposition, dilution or shaping; cadence moved outside the searched family; semantic non-identifiability (Table 4) |
| 3. Active physical authentication | A load behind the meter responded to an unpredictable verifier challenge   | Operator participation; a challenge-bound schedule                    |            | Specified, not run (Appendix H)                                                                                                                                                                                                                                                                             | A challenge-aware dummy load; no semantic binding by power alone                                                                           |
| 4. Work verification              | The physical response is tied to declared training state transitions       | A sound, fresh transcript or proof and an explicit binding assumption |            | Assumed as an ideal functionality (Appendix H)                                                                                                                                                                                                                                                              | The remote-proof / colocation gap, unless the assumed primitive closes it                                                                  |

tion on the observable, not a claim that everything lives in the spectrum; cyclostationary structure is expressed spectrally, but the challenge correlation of the active protocol is a time-domain operation on the same trace. Second, the population: we do not claim a recoverable cadence is universally necessary for training, and some conceivable execution can be made fully aperiodic. We target efficient, synchronous, iteration-structured training that exposes a recoverable cadence, the expected physical signature of that regime where distributed synchronisation makes the modulation deep and coherent. Destroying that cadence is then an adversarial action with a measurable budget, not a counterexample to the method, and measuring that budget is what Section 6 does. The declared workload need not be pure inference either: a facility may be permitted fine-tuning or training below a threshold, whose lines our detector would also find, and telling which line and how large a run it represents needs the integrated-energy channel we exclude. Third, the horizon: the results are fixed-horizon, and we make no claim that our statistics support anytime-valid monitoring, deferring sequential e-process treatment. The cryptographic binding of the active protocol is assumed as an ideal functionality, not built. Real distributed hardware enters only as the falsifiable predictions and the negative transport case of Section 7.2.

**Limitations.** The scenario model is literature-parameterised, not calibrated to measured hardware,

so every quantitative result is a sensitivity analysis over a stated parameter family; the  $\sim 75\times$  modulation gap of Section 7.2 is the standing reminder. The observation map is a linear time-invariant first approximation. The optimality ceiling of Appendix C is optimal only under the Whittle model, hence a lower bound on the true optimum. Every robustness claim is bounded by the adversarial strategies and budgets we specified; a strategy we did not model is outside it. The most important such strategy is energy storage between the accelerators and the meter. A store sized to the cadence swing can discharge during each compute phase and recharge during each communication phase, so that the meter sees a flat load. This costs the operator no throughput, so our frontier cannot price it; its costs are converters rated for the full swing, wear from cycling at the cadence, and conversion losses. Power smoothing of this kind is a grid mitigation for training load in its own right (Ko & Zhu, 2025), so it may be installed for reasons unrelated to concealment. The verifier’s remedy is placement: a meter between any storage and the accelerators sees the unsmoothed load. The tracker’s thresholds are calibrated by Monte Carlo on the modelled null, not per trace, and the known-line veto that would handle instrument lines is specified but not evaluated. Finally, throughput is the only cost we measure. Learning efficiency is argued, not measured, and remains the central open empirical question. It matters directly: the one attack

that weakens the tracker is free in throughput, so learning efficiency is the only place it could be paid for.

**Governance reading.** What a regulator can buy today is a screen. From an external meter alone, meeting the specification of Section 5.2, a Viterbi tracker detects the iteration cadence of synchronous training against a declared inference workload, robust to cadence wander, drift, and moderate variable-work evasion, which already defeats every fixed-frequency test. Extreme variable work halves its detection rate at no throughput cost; whether that attack costs learning efficiency is untested. It does not certify that the computation is training rather than a schedule-matched mimic, benign or deliberate, and it cannot by itself tell training from a periodic instrument line. Both are limits of the passive channel, not of the method, and they run both ways: they cap what a positive detection proves as well as what a counterfeit could fake. A positive detection should therefore trigger escalation — to the active protocol of Appendix H and the cryptographic primitive it assumes — rather than stand as proof of training. The passive meter is a cheap, deployable front line that narrows the question and prices the cheap evasions.

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## A. Fixed-frequency baselines

This appendix gives the two fixed-frequency comparators of Section 4 in full.

When the cadence is stationary, a repeated iteration structure appears as a line, or a harmonic comb, in the power spectrum, and the natural test is the Thomson multitaper harmonic F-test (Thomson, 1982; Percival & Walden, 1993). Taper the mean-removed trace with  $K$  discrete prolate spheroidal sequences  $v_k[n]$  of time–bandwidth product  $NW$ , and let  $y_k(f)$  be the Fourier transform of  $v_k[n] x[n]$  and  $U_k = \sum_n v_k[n]$  the  $k$ -th taper’s gain at zero frequency. The least-squares amplitude of a line at  $f$  and the harmonic F statistic are

$$\hat{\mu}(f) = \frac{\sum_k U_k y_k(f)}{\sum_k U_k^2}, \quad F(f) = (K-1) \frac{|\hat{\mu}(f)|^2 \sum_k U_k^2}{\sum_k |y_k(f) - \hat{\mu}(f) U_k|^2}, \quad (21)$$

the power in the fitted line over the power the tapers see around it. We use  $K = 7$  and  $NW = 4$ , on a zero-padded frequency grid. At a fixed frequency  $f$  the statistic  $F(f)$  is distributed as  $F(2, 2K - 2)$  under a locally smooth Gaussian spectral null with  $K$  tapers, so the test is *approximately pivotal under a stationary, locally smooth spectral null*. It is not an exact false-alarm guarantee across arbitrary external power traces: coloured baselines, controller interference, and co-resident load all violate the local-smoothness premise. Because the operator’s cadence is not known a priori, the test must search a band of candidate frequencies and harmonics, which requires a search correction.

The analytic correction is itself unreliable. Sweeping the padded frequency grid and applying a Bonferroni correction over the Rayleigh-resolution cell count is *anti-conservative by a factor of roughly 3.6–6×*, uniformly across our null suite: the maximum of  $F$  over the padded grid behaves like about five times more effective independent tests than the Rayleigh count, because the  $F$  ratio fluctuates below the Rayleigh scale. The analytic multitaper tail is therefore usable only as a ranking statistic; its operative false-alarm calibration is the Monte-Carlo one supplied by the harness of Section B.3. A spectral matched filter (peak PSD at the searched cadence) serves as the simple comparator. Both are honest baselines for a stationary line, and both fail the moment the line wanders: as the comparison of Section 5.1 shows, the fixed multitaper F-test retains essentially no usable detection power once the cadence drifts, which is exactly why the adaptive setting needs a different detector.

## B. Order-tracked cyclostationary attribution and null validity

The main text detects training with the Viterbi score alone. This appendix reports a longer pipeline that the main text does not rely on: the Viterbi path defines a de-warp of the trace, the de-warped trace is tested for cyclostationarity at the tracked order, and the result is calibrated per trace against a surrogate null. Its value is attribution. Phase-coherent cyclic structure at the tracked order partly separates training from a power-controller limit cycle, which the Viterbi score cannot (Section B.4), but only when thresholds are set against a controller population, and the controller still inflates its surrogate-calibrated false-alarm rate (Section B.3).

### B.1. Tracked cyclostationary detection

A concealing operator breaks the benign baselines cheaply by letting the cycle frequency wander, smearing the line across the searched band. We restore coherence by estimating the wander and undoing it before testing. The detector has three components, each with one job. First, the best path of the Viterbi tracker (Equation (14)), read as a function of time, estimates the instantaneous cycle frequency  $\hat{f}_0(t)$ ; here the tracker is a nuisance-parameter estimator, *not* the detector. Second, the phase estimate

$$\hat{\theta}(t) = 2\pi \int_0^t \hat{f}_0(s) ds \quad (22)$$

defines a resampling of the trace at equal increments of  $\hat{\theta}$  (tachless order tracking (Bonnardot et al., 2005)), so a wandering line becomes a constant-order line. Third, the de-warped signal is tested for cyclostationarity at candidate cycle frequency  $\alpha$  with the Dandawaté–Giannakis statistic (Dandawaté & Giannakis, 1994)

$$Q_\alpha = N \hat{r}_\alpha^H \hat{\Sigma}_\alpha^{-1} \hat{r}_\alpha, \quad (23)$$

where  $\hat{r}_\alpha$  collects the estimated cyclic autocovariances at lags  $\tau$  and  $\hat{\Sigma}_\alpha$  is a consistent estimate of their long-run covariance. For a fixed  $\alpha$ ,  $Q_\alpha$  is asymptotically  $\chi_{2L}^2$  under the structural null, given weak dependence, consistent covariance estimation, and regularity.

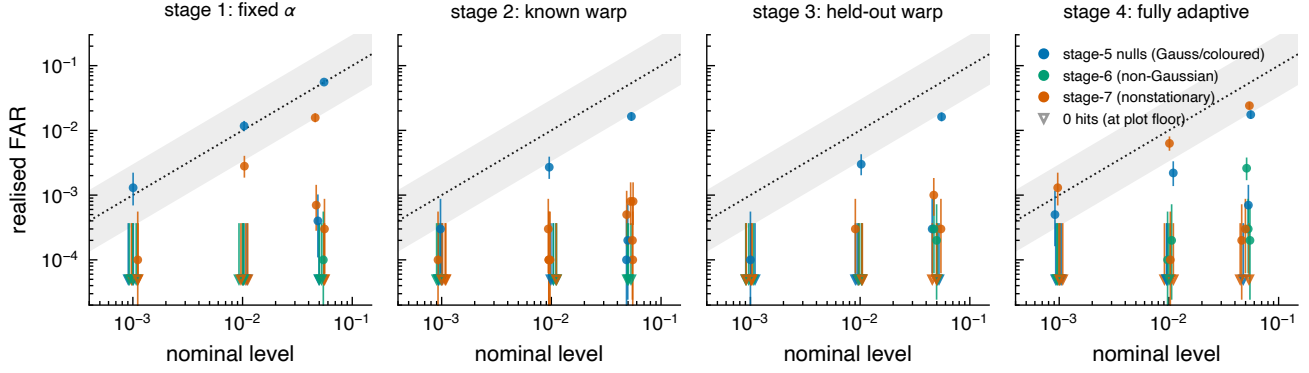


Figure 8. Realised versus nominal false-alarm rate for the adaptive pipeline (Dandawaté–Giannakis with a Bartlett HAC covariance), one panel per stage, every null in the suite, with Clopper–Pearson 95% intervals; nulls are coloured by stage group. No stationary null exceeds the nominal rate at any stage: the two-sided miss is *deflation*, never *inflation*.

Those conditions do not automatically cover the complete adaptive pipeline. The tracker performs an effective search over frequency paths (a look-elsewhere effect); the time warp is estimated from the same record it tests; the covariance estimator  $\hat{\Sigma}_\alpha$  must be chosen; and the asymptotic  $\chi^2$  tail may be a poor guide at finite length. Whether the null survives these steps — and in what form it can be claimed — is the make-or-break question of Section B.3. We do not call the adaptive pipeline an analytic CFAR detector; Section B.3 establishes what it is instead.

## B.2. Surrogate calibration

Because the asymptotic tail cannot be relied on for the complete pipeline — Section B.3 shows why — we calibrate each trace against a surrogate null generated from the trace itself. Given an observed record we form  $S$  surrogates by taking its discrete Fourier transform, replacing each phase with an independent uniform draw while preserving the modulus, and inverting. The resulting series has the observed record’s power spectrum and no phase structure, so the ensemble encodes the invariance “linear and stationary with this spectrum”. The *entire* pipeline — tracker, resampling, and test — is then rerun on each surrogate, and the reported  $p$ -value is the rank of the observed statistic in that ensemble. Running the full pipeline inside the surrogate loop is the point: it is what makes the tracker’s path search and the estimated warp part of the null rather than unmodelled extra freedom.

Two properties of this construction are worth stating in advance, because they determine what the resulting test can and cannot claim. It is exact only against the invariance it encodes, so a null that is genuinely nonlinear or genuinely cyclostationary is not covered — and a controller limit cycle is exactly such a null. And it is expensive: each trace costs  $S$  additional full pipeline evaluations, which is why the surrogate campaigns below are sized smaller than the analytic ones.

## B.3. Null validity of the complete pipeline

The question is whether the adaptive pipeline has a null we can defend. We answer it with a staged design that isolates each adaptive ingredient in turn, crossed with a graded suite of nulls, and measured by the realised false-alarm rate at operationally small nominal levels — not only 0.05. The four stages add one adaptive step at a time: (1) a fixed cyclic frequency  $\alpha$  with no resampling; (2) resampling under a *known* independent time warp; (3) a warp *estimated* on held-out blocks and scored only on the complement; and (4) the fully adaptive full-trace Viterbi tracker. The null suite grades from stationary Gaussian and coloured processes (white, AR(1), a resonant AR(2)), through stationary non-Gaussian processes (heavy-tailed, skewed, squared Gaussian), to locally stationary and controller-like processes (a Van der Pol-type power-management limit cycle mimicking the measured A100 controller). Each cell draws  $M = 10^4$  null traces; realised rates carry Clopper–Pearson 95% intervals. Figure 8 shows the whole grid.

**Level safety holds everywhere.** Across the entire campaign no stationary Gaussian, coloured, or non-Gaussian null inflates the realised rate at any stage or level: the failure mode the gate was built to catch — adaptivity silently raising the false-alarm rate — does not occur. On the white null the fixed- $\alpha$  stage is near nominal (0.056, 0.012, and 0.0013 at nominal 0.05,  $10^{-2}$ , and  $10^{-3}$ ); on coloured and non-Gaussian nulls the same stage is strongly conservative (realised rate  $\approx 0$ ). Tracing the three adaptive steps on the white null, where the fixed- $\alpha$  stage was near nominal, isolates what each one does:

- **Resampling alone (stage 1→2)** deflates the rate by about  $3.4\times$  ( $0.056 \rightarrow 0.016$  at nominal 0.05). Interpolating onto the angle grid correlates adjacent samples, the Bartlett estimator reads that as extra dependence and inflates  $\hat{\Sigma}_\alpha$ , and  $Q_\alpha$  shrinks. Resampling *colours* the series.
- **Estimating the warp (stage 2→3)** moves the null nowhere ( $0.016 \rightarrow 0.016$ ): the held-out design does what it promises.
- **Path selection (stage 3→4)** adds no significant inflation ( $0.016 \rightarrow 0.018$ ; every interval pair overlaps). Even on the resonant AR(2) null, which the Viterbi tracker provably locks onto, the fully adaptive stage fires never in  $10^4$  traces: locking onto a broad noise resonance does not manufacture the phase-coherent cyclic structure  $Q_\alpha$  requires.

The adaptive pipeline is therefore level-safe end to end. What fails is strict two-sided coverage: after resampling the pipeline is uniformly conservative, and the null  $p$ -values are not uniform (Kolmogorov–Smirnov uniformity is rejected at every adaptive cell).

**No analytic calibration closes the deflation.** We swept the open choices — record length, lag set, taper bandwidth, search band, interpolation scheme, and the covariance estimator — selecting on null-calibration quality across the suite, never on power. Only the covariance estimator moves the deflation, and it cannot be closed for the whole ladder at once (Figure 9). Widening the Bartlett bandwidth to  $b = 54$  repairs the adaptive stage two-sided (AR(1) realised 0.050/0.012/0.0016 across the three levels) — confirming that the deflation is an under-bandwidth artefact of the oscillating autocovariance of the resampled series — but the *same* bandwidth breaks the fixed- $\alpha$  anchor, inflating white to 0.073 at nominal 0.05. No single covariance configuration in the swept family calibrates the ladder end to end, and Kolmogorov–Smirnov uniformity is rejected at every adaptive sweep point regardless. The asymptotic  $\chi^2$  tail survives only as a level-safe, power-costing conservative bound; it is not a calibration, and we do not report the adaptive pipeline as an analytic CFAR detector.

**Surrogate calibration recovers the null.** Running the entire pipeline — tracker, resampling, and test — inside a Fourier-phase surrogate null (which preserves the power spectrum and randomises phases, encoding the linear-stationary invariance) recovers exact calibration where the asymptotic tail was  $3\text{--}6\times$  conservative. On the stationary Gaussian nulls the realised rate covers nominal across all three pipeline stages — fixed- $\alpha$ , held-out-warp, and fully adaptive (at nominal  $0.05/10^{-2}$ : white 0.051/0.009, AR(1) 0.050/0.013 fixed- $\alpha$ ; white 0.052/0.013, AR(1) 0.054/0.010 fully adaptive; the resonant-AR(2) and held-out-warp cells likewise) — and Kolmogorov–Smirnov uniformity is not rejected, the property the analytic  $p$ -values never achieved.<sup>1</sup> This is what “calibrated” means throughout the paper: an effective score with per-trace surrogate calibration, not an analytic guarantee.

**The controller marks the attribution boundary.** The surrogate null encodes linear stationarity, so a genuine limit cycle — which is itself cyclostationary — must and does inflate it: on the measured-controller null the surrogate-calibrated rate rises to 0.17–0.30 at nominal 0.05. This is not a defect. The detector correctly reports coherent cyclic structure in a controller trace; calling that structure *training* is the causal-attribution step that needs the nuisance model of Section 3. The controller is the periodic physical confound that Rung 1 alone cannot exclude, and its false-alarm behaviour is exactly the ceiling the claim ladder assigns to this rung.

#### B.4. Power against a periodic confound

The Viterbi score cannot tell a controller limit cycle from training, and the order-tracked statistic partly can. Table 6 shows this for a single training run with no background ( $n = 200$  per class, 300 s traces). Positives are scored against two negative classes, each with its own empirical threshold at false-alarm rate  $10^{-2}$ : the modelled inference null, and a structural population of equal parts AR(1), heavy-tailed AR(1), and the measured controller. With  $n = 200$  negatives the

<sup>1</sup>Frozen on the MATS compute cluster over stages  $1/3/4 \times$  eight nulls at the surrogate sizing  $M = 2000$  null traces  $\times S = 999$  surrogates (the level-safety and bake-off numbers above are at  $M = 10^4$ ; surrogate cells are capped lower because each carries  $S$  inner resamples). This supersedes the earlier partial  $M = 1000/S = 199$  grid. The heavy-tailed but *linear* non-Gaussian nulls stay level-safe (conservative:  $t$ -innovation AR(1) 0.037/0.005, log-normal 0.035/0.006 fully adaptive); the one exception is the *nonlinear* squared-Gaussian null, which the fully adaptive stage over-rejects mildly (0.087/0.018,  $\approx 1.7\times$ , below the pre-registered  $3\times$  threshold). Its cause is the same linear-stationarity assumption analysed next for the controller: a Fourier-phase surrogate cannot reproduce the quadratic phase structure of a squared process and so under-spreads the null. The level-safety claim is thus scoped to linear-stationary nulls.

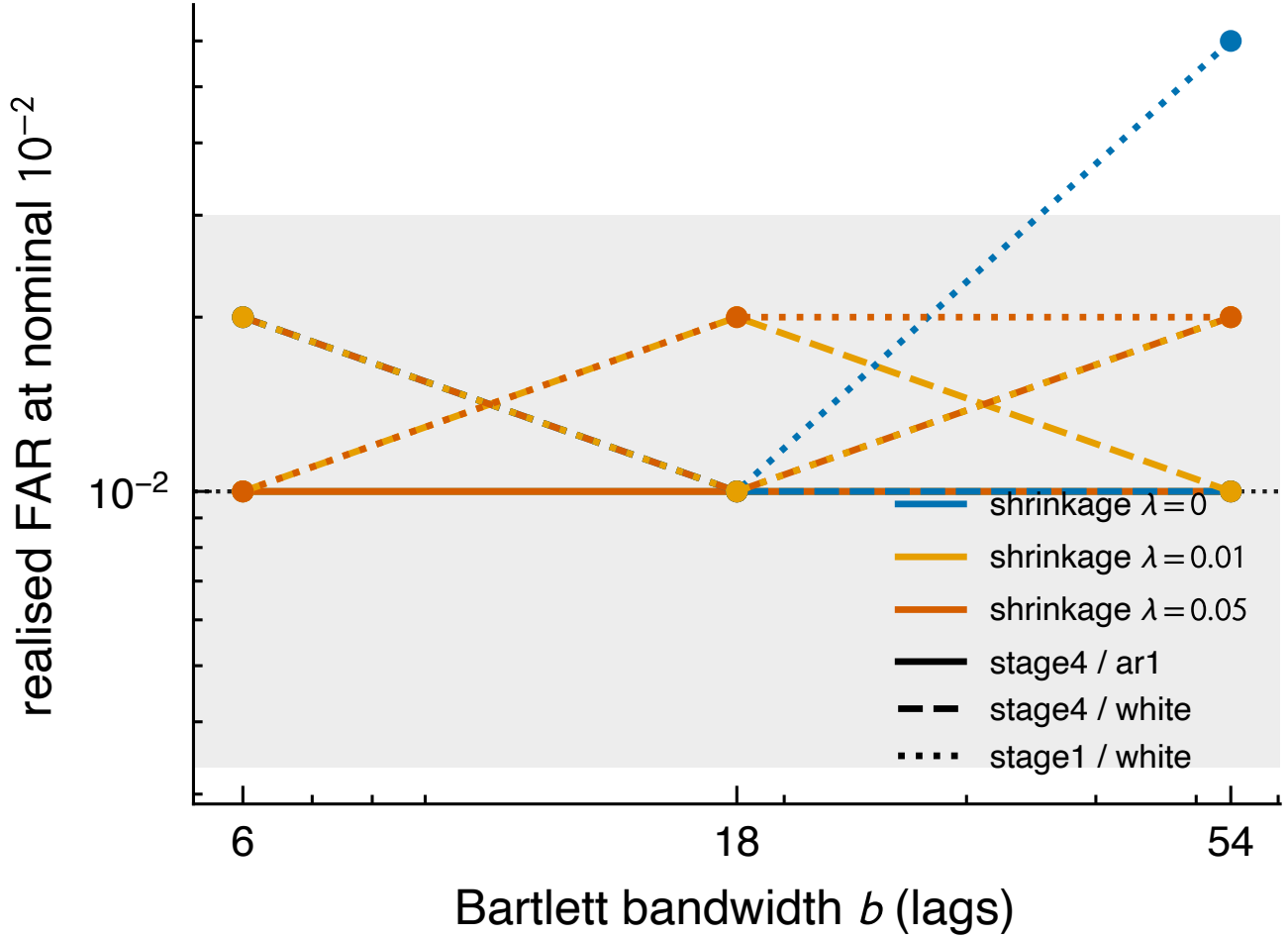


Figure 9. The covariance-estimator sweep is the fix-or-reframe axis: realised rate at nominal  $10^{-2}$  versus Bartlett bandwidth  $b$  for each shrinkage. The bandwidth that repairs the fully adaptive stage (stage 4) inflates the fixed- $\alpha$  anchor (stage 1). No single configuration calibrates both.

$10^{-2}$  threshold rests on the second-highest negative score. Against the inference null the Viterbi score is the strongest detector. Against the structural population it has no power at any drift, because a limit cycle is itself a persistent wandering line and scores as highly as training. The order-tracked statistic keeps power against the structural population at low drift, because only phase coherence at the tracked order distinguishes the two. The sharpest case is controller-only negatives at  $\sigma_f = 0.4$  Hz. There the held-out-warp and fully adaptive variants detect at 0.81 and 0.91 at  $10^{-2}$ , while the Viterbi score detects nothing.<sup>2</sup> The order-tracked power fades with drift and is gone by  $\sigma_f = 0.8$  Hz. This is the separation Rung 1 can deliver against a periodic confound. It needs thresholds set against a controller population, and the scope is a single run: we did not rerun the structural negatives on the aggregate.

### C. Optimality ceiling

Because we own the generators, we can benchmark the tracker against an optimality ceiling: an approximation to the Neyman–Pearson optimal test between the training and null generators. The optimal test compares how likely a trace is under  $\mathcal{H}_1$  and under  $\mathcal{H}_0$ . Our generators can simulate traces from either class but cannot give the probability of a given trace, so we approximate. We summarise each class by its average power spectrum in the cadence band, estimated from simulated traces, and ask which spectrum better explains the periodogram of the trace (a Whittle likelihood ratio). The training cadence is unknown, so the training spectrum is averaged over a bank of 61 per- $f_0$  templates, each estimated from

<sup>2</sup>results/st1/bakeoff.summary.json, including the controller-only hard case.

Table 6. Detection rate at empirical false-alarm rate  $10^{-2}$  by cadence drift  $\sigma_f$  (Hz), for a single training run with no background ( $n = 200$  per class), against the inference null and against the structural population (AR(1), heavy-tailed AR(1), and the measured controller). The Viterbi score cannot separate training from a limit cycle; the order-tracked statistic partly can at low drift.

| Detector                         | 0    | 0.1  | 0.2  | 0.4  | 0.8  | 1.5  |
|----------------------------------|------|------|------|------|------|------|
| <i>vs. inference null</i>        |      |      |      |      |      |      |
| Viterbi                          | 1.00 | 1.00 | 1.00 | 1.00 | 1.00 | 0.98 |
| DG order, held-out warp          | 0.28 | 0.17 | 0.04 | 0.01 | 0.01 | 0.01 |
| DG order, fully adaptive         | 0.73 | 0.62 | 0.55 | 0.23 | 0.06 | 0.01 |
| <i>vs. structural population</i> |      |      |      |      |      |      |
| Viterbi                          | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 | 0.00 |
| DG order, held-out warp          | 0.30 | 0.17 | 0.04 | 0.01 | 0.01 | 0.01 |
| DG order, fully adaptive         | 0.80 | 0.70 | 0.61 | 0.31 | 0.10 | 0.01 |

2000 simulated traces with drift levels drawn from the whole  $\sigma_f$  axis, so the ceiling knows no more about the drift than the tracker does. For the aggregate of Section 5.1 the bank fixes the dominant run’s cadence over a random background, and the null spectrum is estimated from the null aggregate. The simulated traces that build these spectra are drawn from a stream disjoint from the traces we score, and the ceiling is applied to exactly the evaluation populations of Figure 5.<sup>3</sup>

The ceiling is a benchmark, not a usable detector. It is fitted to our own generators, so it knows the exact spectra of both classes, which a verifier monitoring a real facility does not. The tracker assumes only a slowly wandering line, and the ceiling measures how much of the available power it recovers.

The ceiling detects perfectly: detection rate 1.0 at every drift (Figure 5). The training line and the line-free null are therefore separable in principle from a 300 s trace, and the tracker’s shortfall of at most 0.04, at  $\sigma_f = 1.5$  Hz, is the detector’s rather than the problem’s. The natural candidate for closing it comes from the line-tracking literature. Under the hidden-Markov wander model the tracker already assumes, the Viterbi score is the best-single-path approximation, and the Neyman–Pearson statistic is the forward-algorithm marginal over *all* frequency paths (Streit & Barrett, 1990). Evaluating that statistic is future work.

The ceiling is optimal only under the Whittle model. That is a second-order, stationary-Gaussian statistic, which discards harmonic-phase coherence and non-Gaussian structure, so the true optimum can only exceed it. Since the ceiling is already perfect, this changes nothing here.

**The background.** The same comparison on a single training run against inference alone, with no background load, gives tracker detection rates of 1.00 up to  $\sigma_f = 0.8$  Hz and 0.98 at 1.5 Hz, within 0.02 of the aggregate at every drift.<sup>4</sup> At the default shares the background neither helps nor hinders the tracker.

## D. Full attack sweep

The main text reports hiding against budget knob for every schedule strategy, with measured costs where they exist (Figure 7). This appendix gives the full sweep: detection against physical de-periodicisation for every strategy of Table 3 and both detector classes (Figure 10), the per-detector response to variable real work (Figure 11), and the cost-of-hiding view against each detector class (Figure 12). Here the tracking class is the best of the Viterbi tracker and the two order-tracked Dandawaté–Giannakis statistics of Appendix B; on every cell with a measured cost its envelope equals the Viterbi tracker’s alone.

The sweep runs on the aggregate of Section 5, with each strategy applied to the dominant training run only. At every knob setting it also records the physical de-periodicisation the attack induces, from the generator’s ground-truth iteration boundaries: the cadence coefficient of variation, which Figure 10 plots detection against, and the phase-diffusion coefficient. It was frozen on the MATS compute cluster (slurm array 14354, assembled by job 14355):  $n = 200$  traces per class over 50 cells, seed 0, all five detectors, written to `results/st2/aggregate/frontier.summary.json`. Every rate is read at false-alarm rate  $10^{-2}$ . The cost axis reads only the throughput overhead of the single-A100 campaign. It

<sup>3</sup>Frozen on the MATS compute cluster (slurm job 14048); `results/st1/np.ceiling.aggregate.summary.json`.

<sup>4</sup>`results/st1/bakeoff.summary.json` against `bakeoff.aggregate.summary.json`, same seed and  $n = 200$  per class.

ST2 de-periodicisation frontier (% = measured throughput overhead)

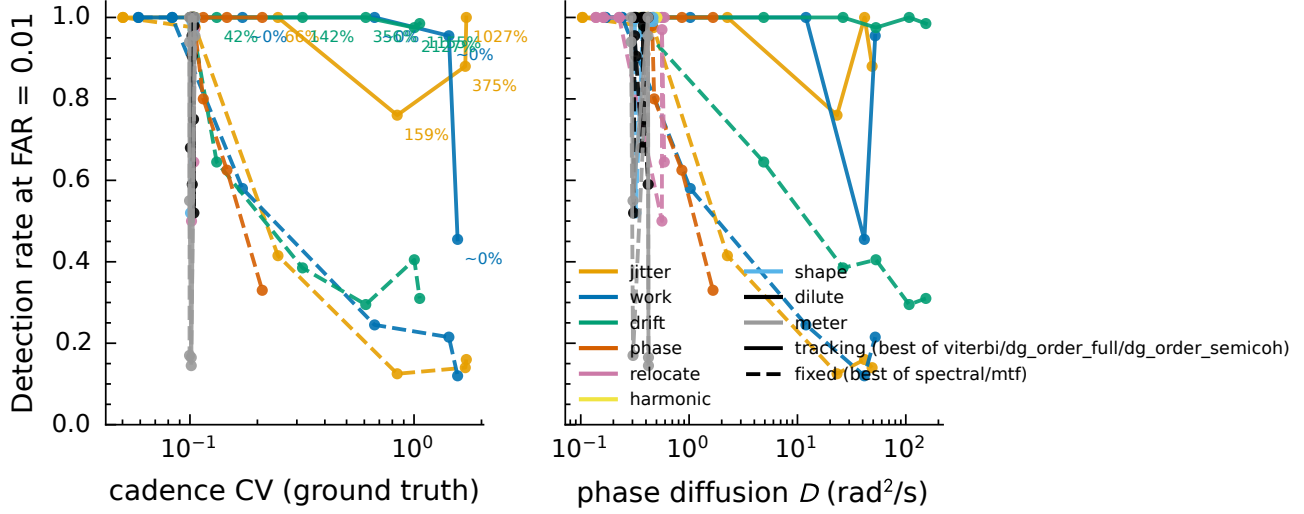


Figure 10. The de-periodicisation frontier: physical de-periodicisation (cadence coefficient of variation) against best tracking and best fixed detection rate at false-alarm rate  $10^{-2}$ , over all adversarial strategies. The near-vertical drop of the fixed class at the variable-real-work strategy, at essentially zero throughput cost, while the tracking class holds, is the scope boundary; costlier strategies (idle jitter, drift, shaping) sit at high throughput overhead without erasing the tracked line.

does not use that campaign’s measured line amplitudes, whose transfer to a facility meter Section 7.2 questions, so the frontier does not depend on that caveat. Each overhead is the mean over eight A100 traces per level. Work variation and idle jitter at level 0.7, and drift at 0.1, 0.4, 0.8 and 1.5 Hz, were added in a later top-up on the same hardware model (data/measured\_cost\_anchors/README.md). The drift overheads are heavy-tailed: at 0.8 and 1.5 Hz the mean exceeds the median by a factor of 1.5 and 2.

**The four strategies without a cost.** Phase slip, relocation, harmonic smoothing and dilution were swept in the same way as the four priced strategies, but the hardware campaign never implemented them, so they have no measured cost and appear in Figure 7 without a cost axis. They do not trouble the tracker at the levels swept: it detects every cell of all four at detection rate 1.0, dilution down to a 0.2 training share included, while the fixed tests fall as low as 0.33 under phase slip (Figure 10). A price for them would therefore only add points at zero hiding from the tracker. For relocation and harmonic smoothing a throughput price would also mislead, because their cost falls mainly on the training dynamics or on energy rather than on throughput.

## E. Meter specification details

This appendix supports the minimum meter specification of Section 5.2 with two results: the notch sub-sweep, and a check that the specification does not depend on the background load.<sup>5</sup>

**The notch.** Figure 13 shows the tracker’s detection rate as a delivery-path notch moves across the cadence band of Table 1 (0.5–1.5 Hz), with the meter otherwise nominal. The notch has quality factor  $Q = 1.5$  (Table 2), so it is broad: about 0.7 Hz wide at 1 Hz. A notch of 50% depth costs nothing at any centre, and one at 0.6 Hz, near the low edge of the band, costs at most 0.02. Deeper notches nearer the band centre cost more. At 0.8 Hz the tracker falls to 0.86 (90% depth) and 0.80 (100%), and at 1.0–1.2 Hz to 0.45–0.63. No notch in the sweep erases detection.

<sup>5</sup>Frozen on the MATS compute cluster (slurm array 14049), 48 cells,  $n = 200$  per class, 300 s traces; results/st2/meter\_boundary\_aggregate\_summary.json. The single-workload sweep is slurm array 5498, results/st2/meter\_boundary\_summary.json.

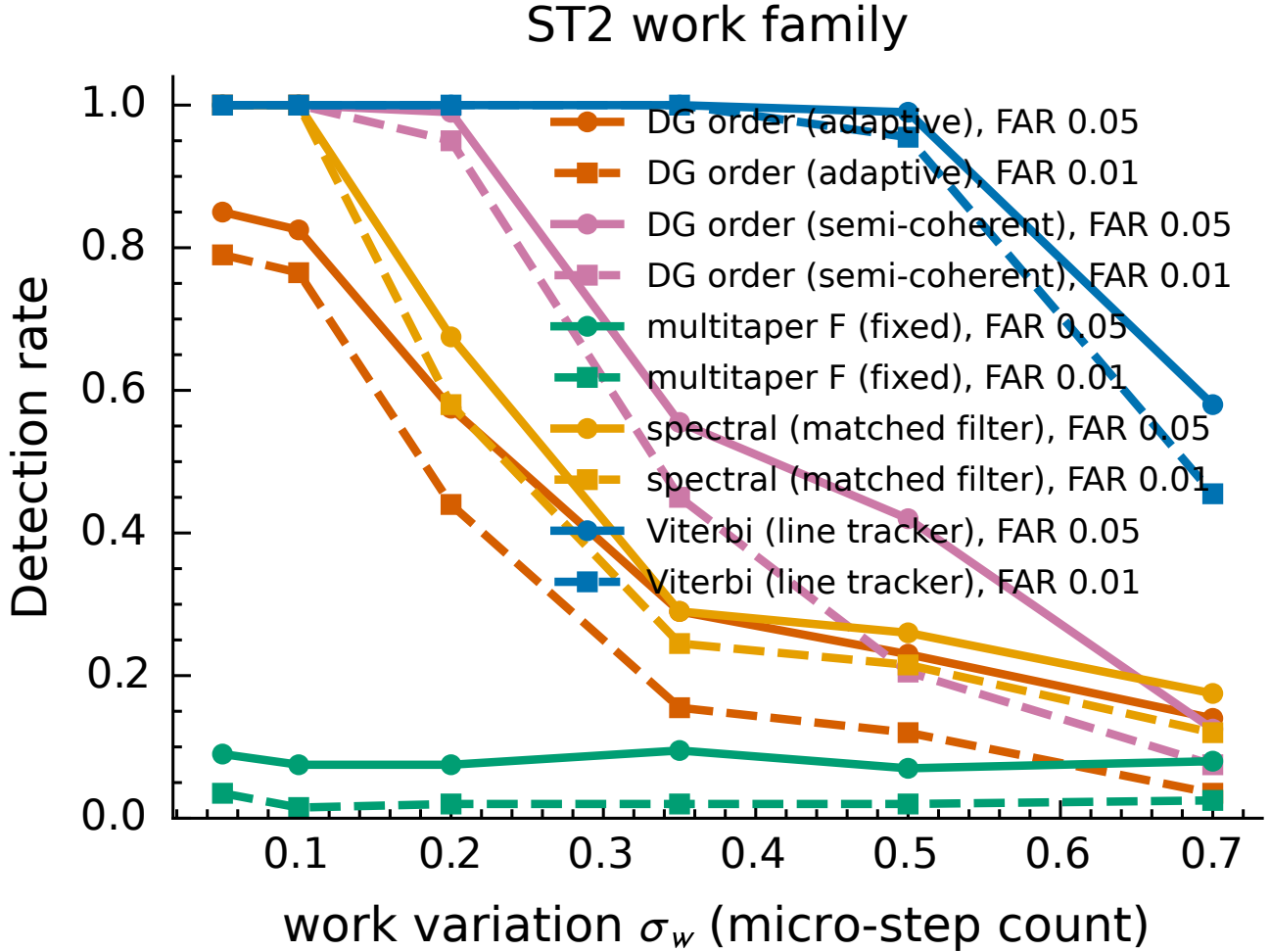


Figure 11. Detection rate versus variable-real-work level for each detector, at false-alarm rate 0.05 (solid) and  $10^{-2}$  (dashed). By level 0.35 the fixed matched filter has fallen below chance (AUC 0.46), and the multitaper F-test stays at its floor throughout; the tracked order statistics degrade more slowly. The Viterbi tracker holds detection rate  $\geq 0.95$  at  $10^{-2}$  until it bends at level 0.7 (detection rate 0.46) — a level whose measured throughput cost is zero within noise, and whose learning cost we do not measure.

**The background.** The same sweep on a single training run against inference alone, with no background load, changes the tracker’s detection rate by at most 0.09 in any cell and leaves the region where detection holds ( $f_s \geq 2$  Hz and  $\tau \leq 0.5$  s) unchanged. As for the drift sweep (Appendix C), the background at the default shares does not move the result.

## F. Conditional classification against the inference null

The main text detects training against the inference null with a single statistic. This appendix reports a fuller classification task built on the same detectors — an eight-feature vector, three decision rules, six domain shifts, and seven semantic controls — whose control outcomes are summarised in Table 4.

### F.1. From detection to classification

Rung 2 sharpens the structural question into a decision: is a trace more training-like than a stated inference population? A decision needs a reference, and ours is the modelled continuous-batching null of Equation (6); every Rung-2 conclusion is conditional on it. We fix the task in advance — the feature vector, the decision rules, and the reporting — so that the result is not retrofitted to the answer.

Each trace is reduced to an eight-dimensional physics feature vector estimated *from the trace alone*, with no oracle knowledge

Cost of hiding: what measured adversary cost buys against each detector class

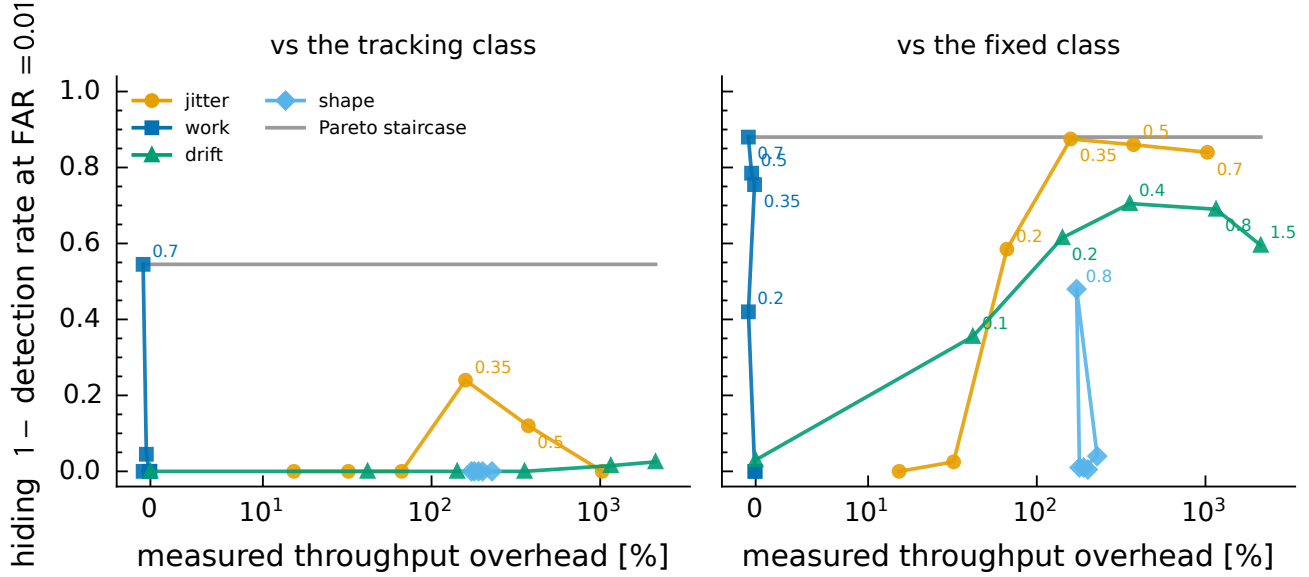


Figure 12. Cost of hiding: measured throughput overhead against hiding ( $1 - \text{detection rate at false-alarm rate } 10^{-2}$ ) for the four priced strategies, with the Pareto staircase — the best hiding available at or below a given cost. Against both classes the staircase is flat from zero cost: variable real work at level 0.7 hides 0.55 from the tracking class (left) and 0.88 from the fixed class (right), and no costly attack does better. The cost axis is linear within  $\pm 10\%$  and logarithmic beyond; the four strategies without a hardware cost are not shown. Knob settings are labelled where they hide something; the rest sit on the floor.

of the cadence: the six Rung-1 detector scores (multitaper F, harmonic comb, spectral matched filter, Viterbi, fixed- $\alpha$  DG, and tracked-order DG), plus two stability features read off the estimated Viterbi path — its frequency-path stability and its phase-folded waveform repeatability. On this fixed vector we compare three decision rules of increasing freedom: (i) a prespecified *physics* score, a fixed linear combination with the fixed-cadence comparators pre-registered out; (ii) a logistic discriminant *fitted* to the scenario generator and scored out-of-fold; and (iii) a random-forest *learned* reference, included to expose how much separation comes from the chosen null rather than from the structural method. We report area under the ROC curve and the false-negative rate at a false-positive rate of 0.05, kept separate throughout from the Rung-1 structural false alarms.

The population the rules are evaluated on is fixed in advance too. They are fitted on the nominal setting and evaluated zero-shot on six domain shifts, each moving one factor: device headroom down to 0.70 and up to 0.95 of peak; observation duration halved to 150 s, which halves the frequency resolution; the cadence band shifted up to 0.8–1.8 Hz, so that its top now lies outside the 0.3–1.7 Hz search band; a meter carrying the controller limit cycle; and a meter with heavy coloured noise and baseline wander.

## F.2. Classification against the inference null

We now apply the three decision rules of Section F.1 to the stated population, the six domain shifts, and a set of semantic controls whose training status we know.<sup>6</sup>

**Stated population and transfer.** On the stated population all three rules separate training from the hard inference null perfectly: area under the curve 1.00 with zero false negatives at a false-positive rate of 0.05 (Figure 14). The separation is not brittle to the population: across all six domain shifts of Section F.1, evaluated zero-shot, the area under the curve stays 1.00. The physical schedule of efficient training is far enough from the inference null that leaving the stated parameter box

<sup>6</sup>Frozen on the MATS compute cluster (slurm array 5612),  $n = 200$  per class, 300 s traces, band 0.3–1.7 Hz, 14 cells (the stated population, six transfer shifts, and seven semantic controls); `results/rung2/rung2-summary.json`.

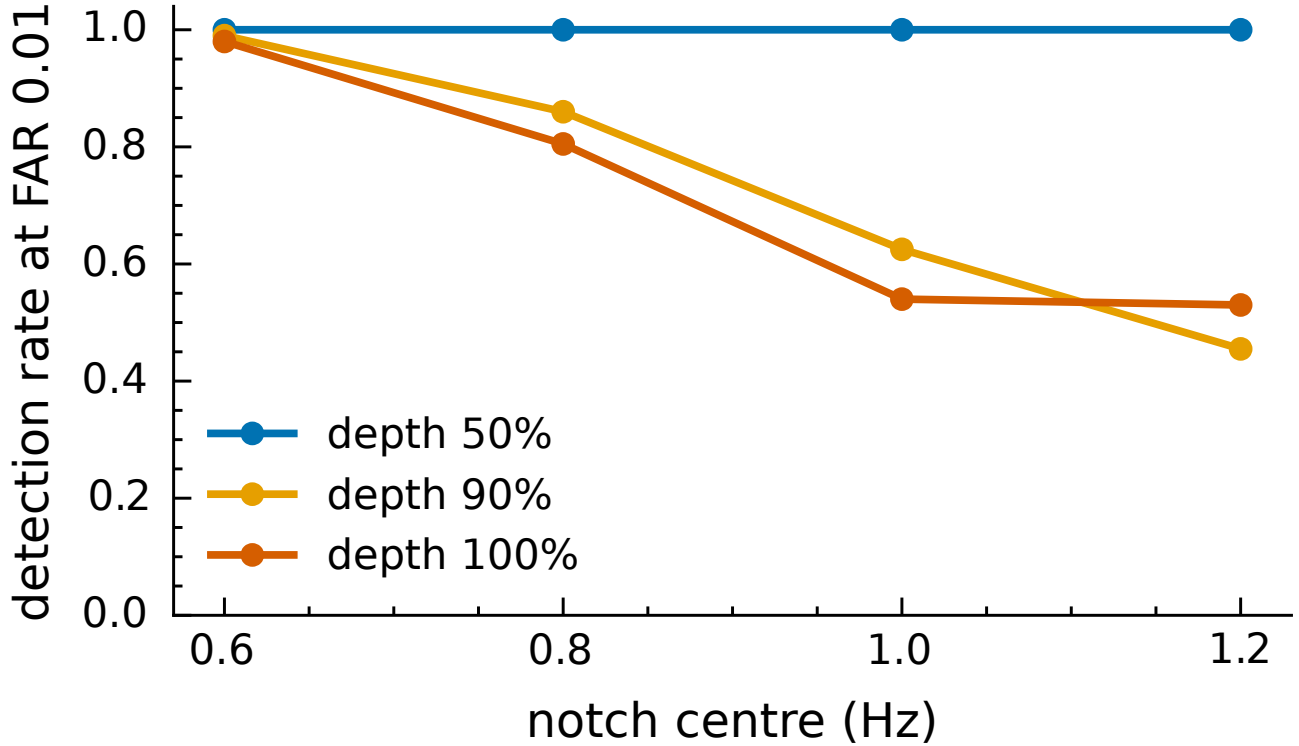


Figure 13. Notch sub-sweep: Viterbi detection rate at empirical false-alarm rate  $10^{-2}$  versus notch centre, one line per notch depth, at 20 Hz sampling with no integration window, the honest aggregate against the null aggregate ( $n = 200$  per class). A deep notch near the band centre roughly halves detection; none erases it. Reproduced by `scripts/plot_st2_meter_boundary.py --scenario aggregate`.

does not close the gap.

**The semantic controls set the ceiling.** The controls are the point of the rung. We built seven loads whose training status we know and asked what each rule scores (Table 4). Five non-training loads — a training-shaped non-ML kernel loop, a controller limit cycle, a periodic inference request generator, a gradient-only computation, and a discarded-update decoy that runs the full forward/backward/optimizer step and throws the update away — *all* score as training, at fraction 1.00 under every rule. A genuine training run diluted by an unrelated co-resident load is still caught (1.00). The reading is the one the claim ladder assigns to this rung: from power alone the verifier can certify a *training-shaped periodic compute schedule*, not that the computation is gradient-descent training rather than a mimic. The first three loads are the practically binding ones, because they need no adversary: the controller cycle scoring as training is the same attribution boundary Rung 1 met in Section B.3, and calling structure training needs the nuisance model of Section 3. The decoy is the extremal member of the same class and the one that cannot be engineered away — Section 7.1 rules out any passive detector that separates it — which is why this ceiling motivates the active protocol of Appendix H rather than a better classifier. The ceiling is a property of the observable rather than of the meter’s bandwidth. A learned classifier reading a board-level probe at 10 MHz separates training from inference by the spectral signature of the memory-bound optimizer-update phase (Gargiulo & Kulp, 2026); the discarded-update decoy runs exactly that phase and discards only its result, so the mechanism such a classifier keys on is present in a load that is not training. More bandwidth resolves the physical schedule more sharply, and it is the schedule, not the semantics, that a passive meter reads.

**Asynchronous training: a robustness result, not a miss.** One control, strongly de-periodicised genuine training, was pre-registered as an expected miss — the case scoped out of Rung 2. At  $n = 200$  it is not missed. The tracked physics rule catches it (area under the curve 0.89, 72% flagged) and the learned reference catches all of it (100%); only the fitted linear discriminant is weak (0.75, 55%). This is consistent with the tracker-robustness story of Rungs 1 and 2: the asynchronous control de-periodicises by drifting the cadence centre and slipping the phase, which leaves a wandering but still coherent

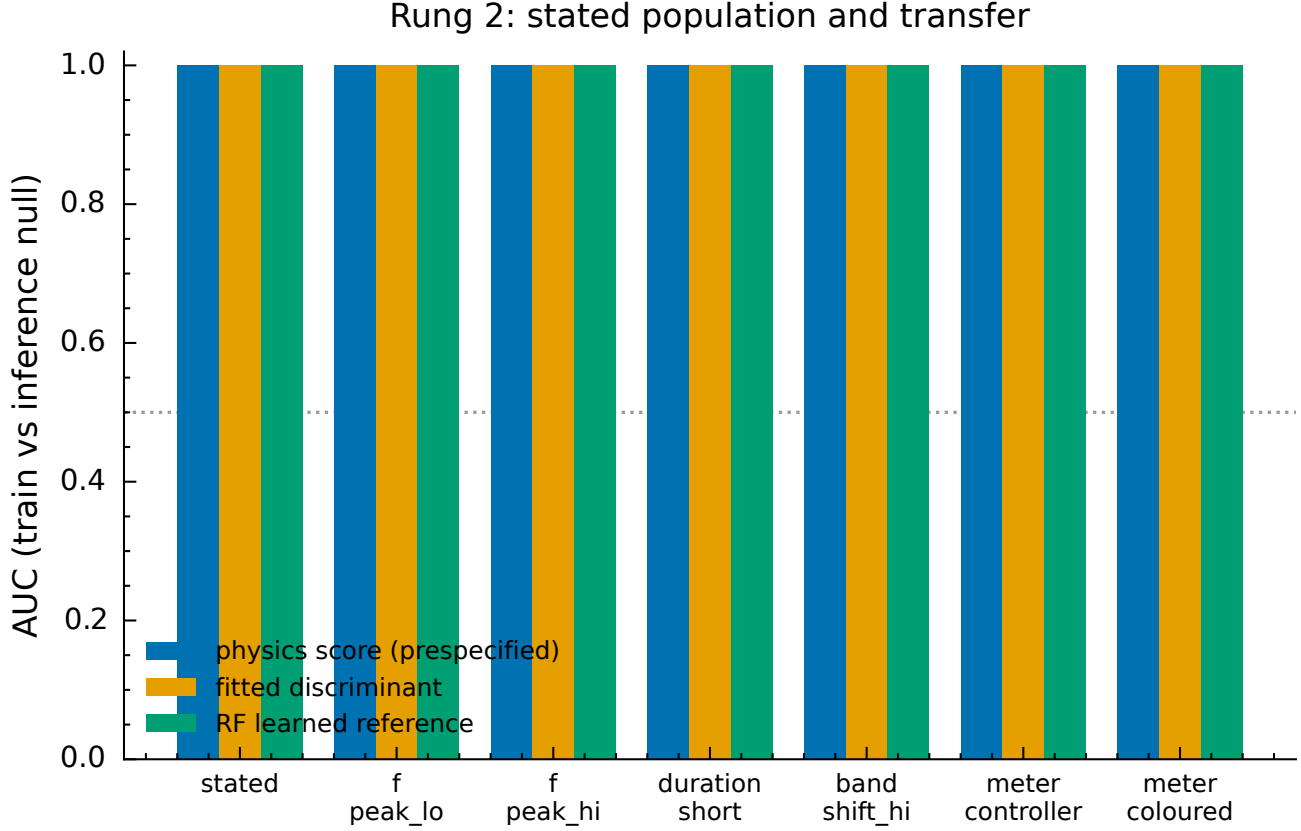


Figure 14. Rung-2 classification on the stated population (ROC / operating points for the three decision rules) and across the transfer shifts. All three rules reach area under the curve 1.00 on the stated population and hold it under every domain shift.

line — exactly the regime order tracking was built to survive. We therefore report it as a robustness result rather than the boundary. The genuine evasion is not slow wander but the  $\approx$ zero-cost variable-real-work attack of Section 6, which moves the cadence *without* leaving a coherent line, and which is where the tracker actually bends.

## G. Identifiability details

This appendix proves the non-identifiability result of Section 7.1.

A passive verifier observes only  $P_{\text{obs}}$ , so its decision is a measurable function  $\phi$  of that trace and its rejection probability is  $\mathbb{E}_{K_h(w)}[\phi]$ . If  $K_{h_0}(w_0) = K_{h_1}(w_1)$  the two expectations coincide for every  $\phi$ , so no test separates the pair, and no post-processing of the trace can (data processing cannot create distinguishability that the laws do not contain). The claim about restrictions follows by contraposition: a non-vacuous guarantee must exclude the equivalent pairs, and each listed lever excludes some of them. An amplitude degeneracy shows the pairs are not exotic. Write the power a workload draws as  $r(t)F(t)$ , where  $F$  is its operation rate and  $r$  the energy cost per operation. The meter constrains only the product, so a workload with a different operation rate and a compensating energy cost per operation is already observationally equivalent, and temporal structure — not amplitude — is what the trace identifies. The near-constructive same-hardware case is the discarded-update decoy: forward, backward, and optimiser kernels executed on the declared schedule with each updated state discarded reproduces the compute/communication alternation, hence the trace, without retaining a model. The converse direction is equally open to the operator: genuine training can be run asynchronously, diluted among other loads, or actively power-shaped, so the equivalence classes are populated from both sides.

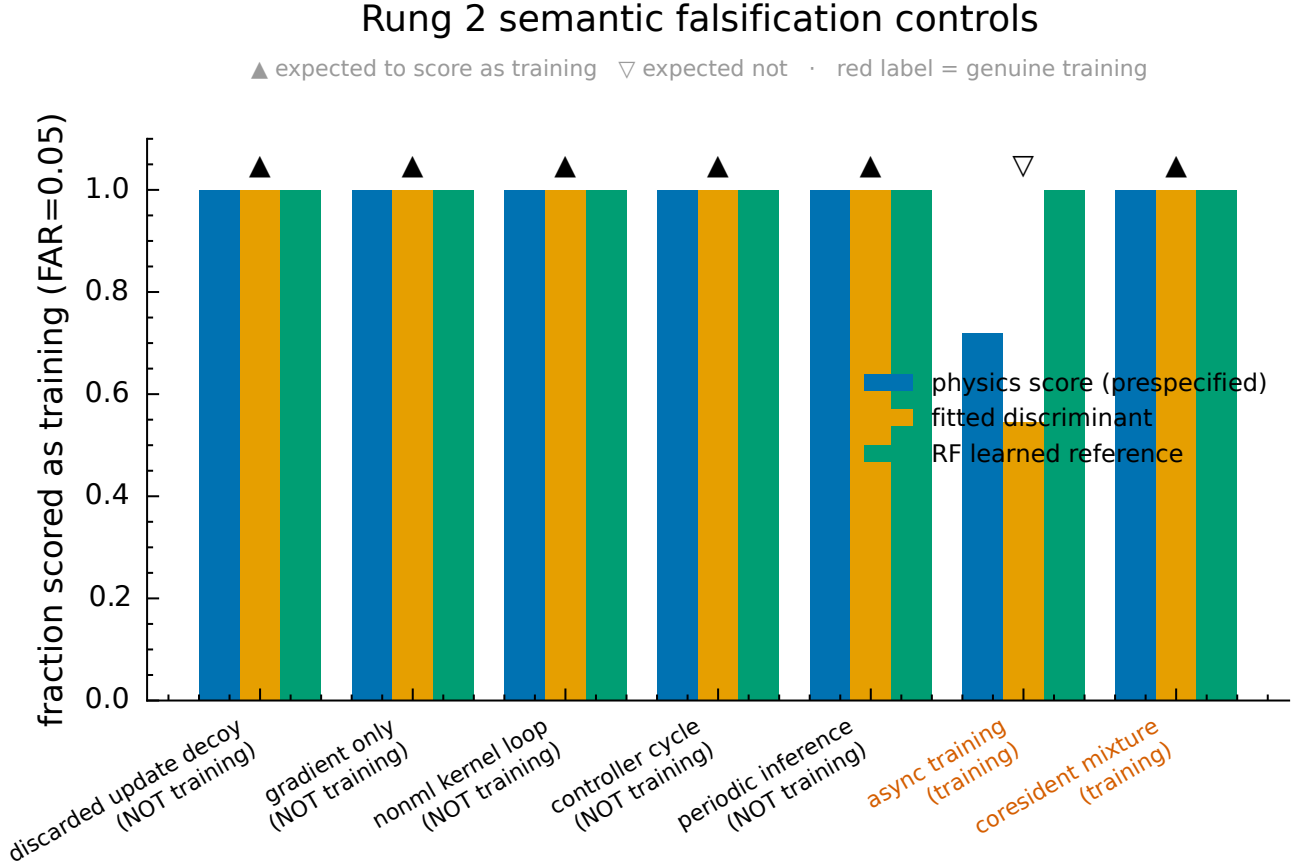


Figure 15. Semantic controls: fraction scored as training per decision rule. The non-training decoys are indistinguishable from training under every rule (the semantic-decoy ceiling); only asynchronous genuine training separates the rules, and the tracked physics rule still catches most of it.

## H. From passive evidence to active authentication (Rungs 3–4)

Rung 2 certifies the physical schedule, not training semantics. Its ceiling is semantic non-identifiability: loads that reproduce the compute–communication cadence of training without training score as training under every passive rule (Section F.2), and no passive test separates them, because the meter sees only the schedule and the mimic copies the schedule. The discarded-update decoy is the sharpest instance — Section 7.1 rules out any passive detector that separates it — even though a concealing operator would never mount one (Section 3.2). The way past a passive ceiling is to stop watching and start asking: authenticate a physical device by its response to an unpredictable query, the premise of distance-bounding protocols (Brands & Chaum, 1994), physical unclonable functions (Pappu et al., 2002) and, in compute governance, of challenge–response telemetry (Monfared et al., 2026). If the verifier perturbs the machine and the machine must react, a mimic that merely replays a recording cannot follow: it does not know what it was asked.

Rung 3 makes this concrete with a challenge-synchronous protocol. The verifier emits an unpredictable balanced challenge sequence  $c_k \in \{-1, +1\}$ ; the operator runs a committed, semantically-equivalent variation of its training schedule keyed to  $c_k$  — for instance a small pre-agreed shift in the boundary between two iterations that leaves the computation performed unchanged. The verifier correlates the external power trace against  $c_k$  over a causal lag window, in the manner of a lock-in amplifier or a spread-spectrum receiver recovering a known code from noise. The null needs no corpus and no matched hardware: it comes from re-randomising the challenge and recomputing the statistic. A positive result authenticates a physical fact — a load behind the meter responded to a signal it could not have predicted. This escapes the semantic decoy, whose passive schedule is by construction uncorrelated with the fresh challenge (Section 3.2). Its own ceiling is the challenge-aware dummy load: a device that receives the challenge and mimics the prescribed response without training. Unlike the passive decoy, this one is an attack an operator has reason to mount, because Rung 3 asks it to demonstrate

training rather than to hide it.

The correlation test is exact under the sharp null that the challenge has no effect on the measured process — but only conditional on the randomisation design, and the guarantee is fragile to the usual abuses (Fisher, 1935). Any maximisation the verifier performs, over causal lags, candidate impulse responses, or preprocessing choices, must be repeated inside every re-randomised replicate, or the reported level is optimistic. Carryover between challenge periods, any adaptivity in the operator’s schedule choices, and balance constraints on the challenge sequence all belong in the design that the re-randomisation samples from. We also flag the weaker reading a careless protocol invites: over a long enough record a correlation detector accrues the  $\sqrt{N}$  processing gain of any matched receiver, so a positive result must be shown to certify the challenge binding and not merely that the verifier integrated for long enough.

Rung 3 authenticates a physical response; it does not tie that response to training. Rung 4 adds that binding, and it is the rung this paper does not build. Rather than gesture at “a proof of training,” we state the missing piece as an ideal functionality (Canetti, 2001): a sound, fresh, challenge-bound transcript or proof that ties each challenge bit to the prescribed semantically-equivalent variation of declared training state transitions. Such a primitive must provide commitment before the challenge is revealed, unpredictability at commitment time, sound binding of the response to valid declared state transitions, freshness against replay, and an explicit position on whether the responding load and the declared computation are co-located. Proof-of-learning constructions (Jia et al., 2021) are the closest existing candidate, but they assume access to the training trace rather than an external meter. Conditional on such a primitive, challenge-synchronous power supplies physical evidence for the declared response; without it, the semantic work binding remains an assumed, open layer.

Two ceilings therefore remain. Rung 3’s is the challenge-aware dummy load; Rung 4’s is co-location: a genuine remote proof combined with a local challenge-correlated dummy still satisfies both the correlation test and the transcript unless the assumed primitive explicitly rules the separation out. Closing them empirically is future work, and it needs no cluster — challenge modulation is legible on a single card. A full Rung-3 study would fix a concrete pair of semantically-equivalent schedules, run the complete search inside every re-randomised replicate, pit the detector against a challenge-aware dummy-load adversary, and trace detection against modulation depth, throughput overhead, and observation time. We leave it as the active-verification direction the passive results here motivate but cannot reach — the response to the ceiling that Section 7.1 makes unavoidable for any passive detector (Section 7).

## I. Scratch

Material cut from the body during the 2026-09 restructure and parked here pending a decision. **This appendix is not part of the paper.**

### I.1. Contributions list (cut from the introduction)

The introduction previously closed with an enumerated contributions list. It was cut because the section it headed is now prose; the claims themselves are made and evidenced in the body.

1. **The claim ladder** (Table 5) as an organising frame for passive compute verification: four claims, their information costs, and their ceilings, closed out one by one in Table 8.
2. **A tracked cyclostationary detector for Rung 1** (Section 4), with a negative methodological result we report rather than bury: no covariance configuration calibrates the complete tracker–resample–test pipeline end to end, so it is not a constant false-alarm-rate test and we do not present it as one. It is an effective score with per-trace surrogate calibration, level-exact on the linear-stationary nulls we tested. A pre-registered bake-off (Section B.4) makes the choice of detector an outcome, and against a Whittle likelihood-ratio ceiling the tracker is essentially optimal on the inference null while only the cyclostationary order family approaches the ceiling on structural confusers.
3. **The de-periodicisation frontier** (Section 6), the central result, priced against measured throughput anchors. Its shape is an asymmetry: against fixed-frequency tests a prover reaches a hiding rate of 0.74 at essentially zero throughput cost by varying real work per iteration, whereas against the tracking class the best anchored attack reaches only 0.16 at 159% overhead.
4. **A minimum meter specification** (Section 5.2): the observation channel is a first-class condition on every rung, and we locate where each detector class dies in sample rate, integration window, and in-band transfer function. A 1 s-integrating, 1 Hz-reporting meter defeats every detector on an unmodified honest workload.

5. **Rung-2 classification with semantic falsification controls** (Section F.2). Three decision rules separate training from the stated inference null perfectly and survive six domain shifts; but five non-training loads — three of them benign operational loads, alongside a decoy that computes gradients and discards them — all score as training. The meter certifies the physical schedule, not training semantics: a ceiling no better passive detector removes, and one that limits positive evidence as much as it admits counterfeits.
6. **A negative transport case** (Section 7.2): what we measured on real hardware, why it does not carry the distributed claim, and what would falsify the prediction that it should.

Supporting these, Section 7.1 gives the formal counterpart of the Rung-1 and Rung-2 ceilings: an observational-equivalence argument for why no passive test can identify training semantics, and a covertness cost bound showing that a prover using i.i.d. phase jitter to reach a target covertness level against an exhibited verifier must pay in throughput or learning efficiency. We are explicit about what that bound does not cover, and Section 6 supplies the near-free attack that shows why the gap matters.

## J. Scratch

**[Parked] Old Detection-methods introduction.** The weakest question on the ladder is whether the trace carries order-coherent power variation that a stated structural null does not produce. Answering it establishes *coherent repeated power structure*, not training: reading that structure as repeated multi-phase computation additionally requires the nuisance model of Section 3 that excludes periodic physical confounds, of which the measured A100 power-controller oscillation (Section 7.2) is the standing counterexample. This section develops the detectors and the calibration machinery; Section 5 reports what they do.

Two prover settings drive the design. A benign prover’s cadence is stationary, and a stationary line has a classical test. An adaptive prover lets the cadence wander, which breaks that test cheaply and forces a different construction. The benign case is easy and the paper is not about the benign case; the adaptive case is the methodological heart, and the burden here is to give its detector a defensible null.

**[Parked] Old main-text detector-power subsection.**

**[Old heading] Detector power and how far it sits below optimal** Which detector to use is an *outcome* here, not an assumption. Before scoring any signal we fixed the detector set, the conditions, and the metric, and ran a pre-registered comparison across a cadence-drift axis  $\sigma_f \in \{0, \dots, 1.5\}$  Hz against two negative classes: the modelled inference null, and a mixed structural population of AR(1), heavy-tailed AR(1), and the measured controller. Section B.4 reports the full grid; three findings from it carry into everything below.

Tracking is necessary. The fixed- $\alpha$  Dandawaté–Giannakis statistic, handed the *true*  $f_0$  as an oracle, collapses from detection rate 0.34 at zero drift to  $\leq 0.1$  under any wander — even the true frequency is useless to a coherent fixed- $\alpha$  test once the line moves — and the fixed multitaper F-test has essentially no usable power against either negative class at empirical thresholds. Only the tracked order family separates the wandering line from structural confounds: against the mixed structural negatives the matched filter and the raw Viterbi score have detection rate zero at every drift, because the controller third of the mixture fires them harder than the signal does. Phase-coherent cyclic structure at the tracked order is the one signature the confuser cannot fake, and this is the attribution boundary of Rung 1 drawn with power rather than with false-alarm rate. The honest limit is fast wander: beyond drift 0.8 Hz every calibratable detector fades, and a 300 s record simply does not resolve fast cadence wander against an honest null.

That ranks the detectors against each other but says nothing about how good the winner is. Because we own the generators we can also benchmark against an optimality ceiling. No closed-form likelihood exists — both classes are black-box samplers — so we use a Whittle spectral likelihood ratio on the in-band periodogram with a Monte-Carlo training template bank marginalised over cadence, fitted on a disjoint stream and applied to the exact evaluation populations. The ceiling is perfect: area under the curve and detection rate are 1.0 in every column at every drift. The training comb and the line-free nulls are separable in principle from a 300 s trace, so all the difficulty the comparison exposes lives in the detector, not in the problem.

Read as a fraction of that ceiling, the picture sharpens in a way the frontier of Section 6 depends on. Against the inference null the Viterbi tracker is essentially Neyman–Pearson optimal, holding the ceiling’s power across the whole wander axis,

Table 7. The claim ladder. Each rung answers a strictly more governance-relevant question at a stated cost in verifier information; each rung’s hard ceiling motivates the next. Within each rung we consider a benign (non-adaptive) and an adaptive (hiding) prover. Table 8 closes each rung out against the evidence.

| Rung                              | Claim                                                                    | Additional requirement                                             | Hard ceiling                                                                                                                                                                               |
|-----------------------------------|--------------------------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Structural evidence            | Order-coherent power variation exists under a specified structural null  | External time-resolved trace only                                  | Periodic physical confounds; structural-null misspecification; strong superposition or dilution; active power shaping; cadence moved outside the searched family; die-to-meter attenuation |
| 2. Conditional classification     | The trace is more training-like than the stated inference population     | A representative or explicitly modelled inference null             | Domain shift; semantic non-identifiability (training-shaped execution, benign or deliberate, without training)                                                                             |
| 3. Active physical authentication | A load behind the meter responded to an unpredictable verifier challenge | Prover participation; a challenge-bound schedule                   | A challenge-aware dummy load; no semantic binding by power alone                                                                                                                           |
| 4. Work verification              | The physical response is tied to declared training state transitions     | A sound, fresh transcript/proof and an explicit binding assumption | The remote-proof / co-location gap, unless the assumed primitive closes it                                                                                                                 |

while the fixed matched filter’s fraction of attainable power collapses from 1.0 at zero drift through 0.78 and 0.38 to below zero by drift 0.4 Hz. Against the structural confusers the tracker is *not* enough: only the tracked order family approaches the ceiling, reaching 0.93 of it at low drift and fading to 0.10 at the fastest wander. The widening gap at high drift is honest headroom above today’s detectors rather than a limit of the channel, and the one caveat runs in our favour — the Whittle ceiling discards harmonic-phase coherence and non-Gaussian structure, so the true optimum can only exceed it.

**[Parked] Old Discussion opening, ladder lead-in, ladder tables, and governance readings.** We set out to ask what a passive power meter can certify about AI training, and organised the answer as a claim ladder in which each rung answers a more governance-relevant question at a higher cost in verifier information, and each rung’s ceiling motivates the next. Table 8 closes the ladder out with the paper’s evidence attached to each rung.

This section fixes what the verifier sees, who is trying to hide from it, and what can be argued about the channel before any detector is run. It also fixes the vocabulary for the claims a meter can support. We organise those claims as a *claim ladder* (Table 5): four claims of increasing governance relevance, each bought at a stated cost in verifier information, and each with a hard ceiling that motivates the next. Rung 1 establishes that order-coherent power structure is present, which is evidence rather than identification. Rung 2 converts that structure into training-likeness at the price of committing to an inference null; its ceiling is semantic non-identifiability. Rungs 3 and 4 leave the passive channel for an active protocol and a transcript primitive, and are specified but not run (Appendix H). The ladder is a ladder of claims and assumptions, not a quantitative bound on unidentified compute. Within every rung we consider two provers: a **benign** (non-adaptive) prover who runs its workload without regard to the meter, and an **adaptive** (hiding) prover who knows the detector and reshapes its schedule to evade it. Most of what is interesting lies in the gap between them.

**Governance reading.** A passive meter certifies efficient iteration-structured training against cheap, fixed tests only within a bounded de-periodicisation budget; a prover free to vary real work escapes those tests for nothing. Defeating the order tracker on top of that requires either a systems and learning cost we could not measure or a degradation of the meter channel itself (Section 5.2). The results of this section are therefore necessary conditions and an upper bound on hideability rather than a proof of training: they bound what a prover can hide from a fixed detector cheaply. The complementary direction — a lower bound on what hiding must cost, which no detection sweep can supply — is available for one adversarial strategy against a fixed verifier (Section 7.1) under the cost-of-hiding assumption, and the distortion it requires can be priced against the same anchors used here. The tracking class dominates the fixed class at every cell we measured, so that threshold is a valid — but loose — reading of what a tracker would require: it is read off the weaker detector, so it understates the true requirement. Establishing it against the tracking class directly, rather than inheriting it from the measured dominance, remains open.

Table 8. The claim ladder closed out: what this paper establishes at each rung, where, and the hard ceiling it hits. Mirrors Table 7.

| Rung                          | What the paper establishes                                                                                                                                                                     | Evidence                | Demonstrated ceiling                                                                                                                                                                                                  |
|-------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Structural evidence        | Order-tracked cyclic structure is detected against stationary and controller-like nulls; the tracker is Neyman–Pearson optimal against the inference null                                      | Section 5, Section B.4  | Fast cadence wander ( $>0.8$ Hz on 300 s); attributing structure to training needs the nuisance model; the meter must meet Section 5.2                                                                                |
| 2. Conditional classification | The trace is more training-like than the stated inference null (area under the curve 1.00, stable across domain shift)                                                                         | Section F.2             | Semantic non-identifiability: non-training loads with a training-shaped schedule score as training, benign ones included; Section 7.1 shows no passive detector can do better                                         |
| Frontier                      | Fixed tests collapse under $\approx$ zero-cost variable real work while the tracker holds up to level 0.5; costly de-periodicisations buy no concealment                                       | Section 6               | At level 0.7 the tracker detects about half of runs at no throughput cost; the learning cost is unmeasured                                                                                                            |
| Cost of hiding                | Hiding to detectability $\varepsilon$ requires a distortion $\sigma^*(\varepsilon)$ that the measured anchors price at tens of percent of throughput, rising above 100% (i.i.d. period jitter) | Section 7.1, Appendix G | Read off a fixed-bin verifier, so valid against any class containing it, but loose; one adversarial strategy; internal to the stated generator; the throughput-or-learning disjunction is an assumption, not a result |
| 3–4. Active authentication    | A challenge-synchronous protocol and an assumed binding primitive, specified but not run                                                                                                       | Appendix H              | Challenge-aware dummy load; the co-location gap                                                                                                                                                                       |

**Governance reading.** What a regulator can buy today is graded. Rung 1 buys physical evidence of an efficient iteration-structured compute schedule from an external meter alone, robust to a wander-and-drift adversary but not to a prover who varies real work or degrades the meter. Rung 2 buys the further claim that the schedule is more training-like than a stated inference baseline — but not that the computation is training rather than a schedule-matched mimic, benign or deliberate, which is a limit of the passive channel, not of the method. That limit runs both ways: it caps what a positive detection proves as well as what a counterfeit could fake. Separating training from a mimic, and binding a physical response to declared work, both require moving off the passive channel to the active protocol of Appendix H and the cryptographic primitive it assumes. The passive meter is thus a cheap, deployable front line that narrows the question and prices the cheap evasions, not a stand-alone proof of training.

The problem of

The problem of X is well established

Inferring which activities are running from one aggregate meter is the founding problem of non-intrusive load monitoring (Hart, 1992; Zoha et al., 2012), with a power-side-channel malware detector as its closest security-flavoured descendant (Clark et al., 2013). Deciding whether a source is active against a transmitter trying to stay below the detection floor is spectrum sensing, whose taxonomy of energy, matched-filter and cyclostationary detectors structures our own comparison (Yücek & Arslan, 2009).

Tracking a narrowbandline whose frequency wanders is passive sonar’s lofargram problem (Streit & Barrett, 1990), carried into continuous-gravitational-wave searches (Suvorova et al., 2016; Bayley et al., 2019), and undoing that wander before testing is tachless order tracking (Bonnardot et al., 2005).

Each of these literatures assumes something the governance setting removes:

load monitoring assumes indifferent appliances with clean switching edges;

side-channel analysis, from differential power analysis onward (Kocher et al., 1999; Chari et al., 2002), assumes a proximate high-bandwidth probe and a cooperative target;

the line trackers assume a stationary noise floor with no adversary behind it.

None of them supplies a calibrated test for a quasi-periodic, frequency-wandering signature observed through a low-rate

filtered meter, and none prices what it costs the source to remove it.

The adversarial half does have an owner. Pricing concealment against detectability is the warden-versus-covert-transmitter game of covert communication (Bash et al., 2013) and the relative-entropy security measure of information-theoretic steganography (Cachin, 2004): both bound how detectable a hidden activity remains as a function of how constrained it is, and both do so for *every* verifier at once rather than for one detector. Their results assume a transmitter free to shape its signal. A training schedule is not free: its communication phase has a physical duration that does not shrink when the compute phase does, and each distortion of the cadence costs throughput or learning efficiency. That constraint is what makes a cost-of-hiding bound possible here, and no existing bound covers it.

What is missing is a method that turns a low-rate external power trace into a calibrated test for iteration-structured training, prices its evasion against an adversary who knows the test, and states what the channel cannot certify at any price. Without one, “power monitoring” enters policy proposals as a binary capability: no one can write the meter specification a regulator would have to procure, budget the evasions a prover would actually attempt, or say which governance claims a positive detection does and does not support.

and U.S. Executive Order 14110 briefly operationalised one, requiring reports of runs above  $10^{26}$  FLOP (The White House, 2023).

Every such scheme presupposes that a regulator can tell when training is happening on monitored hardware. Three channels could tell it. Prover-owned telemetry is informative but conditional on tamper-resistant read paths that do not yet exist (Rahman & Tajdari, 2026; O’Gara et al., 2025); challenge–response protocols remove that trust anchor but require the prover’s participation (Monfared et al., 2026); and a verifier-owned probe on an accelerator’s supply conductors requires neither, at the price of physical access to the board and continuous on-site presence (Gargiulo & Kulp, 2026).

Multi GPU synchronous AI training of the type that typifies frontier AI models alternates a compute phase, a forward backward pass with high power requirements, with a communication gradient all reduce phase where the GPUs all sync and power requirements drop. [what about async training? maybe we need to mention this somewhere? perhaps in limitations? ]

In practice the problem is harder as the probe is at a higher level i.e. more remote a meter three to four orders of magnitude slower than the probe, behind a power-delivery path that low-passes and dilutes the signal, reading a prover who knows it is being read.